CSES Introductory Problems — Problem Reference

All listed tasks: 1 s, 512 MB. Examples omitted for compactness.

1072 — Two Knights

For every $k = 1, 2, \ldots, n$, count ways to place two knights on a $k \times k$ chessboard so that they do not attack each other. **In:** `n`
Out: n integers: the answer for each $k = 1, \ldots, n$
Constraints: $1 \le n \le 10^4$.

1618 — Trailing Zeros

Given n, find the number of trailing zeros in $n!$. **In:** `n`
Out: number of trailing zeros in $n!$
Constraints: $1 \le n \le 10^9$.

1622 — Creating Strings

Given a lowercase string, generate every distinct permutation of its characters. **In:** one lowercase string s
Out: first the number k of distinct strings; then all k strings in lexicographic order
Constraints: $1 \le |s| \le 8$.

1623 — Apple Division

There are n apples with weights p_i. Partition all apples into two groups, minimizing the absolute difference between the groups' total weights. **In:** `n`; then `p_1 ... p_n`
Out: minimum possible weight difference
Constraints: $1 \le n \le 20$, $1 \le p_i \le 10^9$. Use 64-bit.

1624 — Chessboard and Queens

Place eight queens on an 8×8 board so no two attack each other. Queens may be placed only on free squares; reserved squares do *not* block attacks. **In:** 8 rows of 8 characters: `.` free, `*` reserved
Out: number of valid placements
Constraints: Fixed 8×8 board and exactly 8 queens.

1625 — Grid Paths

A path on a 7×7 grid starts at the upper-left square and ends at the lower-left square, visiting each square at most once. Its 48 moves are described by `D`, `U`, `L`, `R`, or `?` (any direction). Count paths matching the description. **In:** one 48-character string over `?DULR`
Out: number of matching paths
Constraints: Grid is fixed at 7×7; description length is 48.

1743 — String Reorder

Rearrange the characters of an uppercase string so no two adjacent characters are equal. Among all valid rearrangements, find the lexicographically smallest. **In:** one string s over `A–Z`
Out: lexicographically smallest valid rearrangement, or `-1` if impossible
Constraints: $1 \le |s| \le 10^6$.

2165 — Tower of Hanoi

Move n disks from stack 1 (left) to stack 3 (right), using stack 2. Move only the top disk of a stack; never place a larger disk on a smaller one. Minimize the number of moves. **In:** `n`
Out: minimum move count k, then k lines `a b` describing each move
Constraints: $1 \le n \le 16$.

2205 — Gray Code

Construct a Gray code of length n: all 2^n bit strings of length n, ordered so consecutive strings differ in exactly one bit. **In:** `n`
Out: 2^n lines forming any valid Gray code
Constraints: $1 \le n \le 16$.

2431 — Digit Queries

Consider the infinite string $123456789101112\ldots$. For each query, find the digit at 1-indexed position k. **In:** `q`; then q values `k`
Out: one digit per query
Constraints: $1 \le q \le 1000$, $1 \le k \le 10^{18}$.

3217 — Knight Moves Grid

A knight is on an $n \times n$ chessboard. For every square, determine the minimum number of knight moves needed to reach the top-left square. **In:** `n`
Out: $n \times n$ matrix of minimum move counts
Constraints: $4 \le n \le 1000$.

3311 — Grid Coloring I

An $n \times m$ grid contains `A`, `B`, `C`, or `D`. Change *every* cell to one of these four characters, different from its original character, so no horizontally or vertically adjacent cells are equal. **In:** `n m`; then n grid rows
Out: any valid final grid; if none exists, `IMPOSSIBLE`
Constraints: $1 \le n, m \le 500$.

3399 — Raab Game I

Each of two players has cards $1, 2, \ldots, n$ and plays each card exactly once. Each round, the higher card scores one point; equal cards score neither. Given final scores a, b, construct a possible play order for both players. **In:** `t`; then t cases `n a b`
Out: for each case, `NO` if impossible; otherwise `YES` and two permutations of $1, \ldots, n$ realizing the scores
Constraints: $1 \le t \le 1000$, $1 \le n \le 100$, $0 \le a, b \le n$.

3419 — Mex Grid Construction

Construct an $n \times n$ grid. Each cell must contain the smallest nonnegative integer that appears neither to its left in the same row nor above it in the same column. **In:** `n`
Out: the required $n \times n$ grid
Constraints: $1 \le n \le 100$.

CSES Sorting and Searching — Problem Reference

All listed tasks: 1 s, 512 MB. Examples omitted for compactness.

1073 — Towers

Process n cubes in the given order. A cube may start a new tower or be placed on a tower whose current top is strictly larger. Minimize the number of towers. **In:** `n`; then `k_1 ... k_n`

Out: minimum number of towers

Constraints: $1 \le n \le 2 \cdot 10^5$, $1 \le k_i \le 10^9$.

1074 — Stick Lengths

Change every stick to one common length. Changing length p_i to x costs $|p_i - x|$. Minimize total cost. **In:** `n`; then `p_1 ... p_n`

Out: minimum total cost

Constraints: $1 \le n \le 2 \cdot 10^5$, $1 \le p_i \le 10^9$. Use 64-bit.

1085 — Array Division

Divide an array of positive integers into exactly k contiguous nonempty subarrays. Minimize the largest subarray sum. **In:** `n k`; then `x_1 ... x_n`

Out: minimum possible value of the maximum part sum

Constraints: $1 \le k \le n \le 2 \cdot 10^5$, $1 \le x_i \le 10^9$. Use 64-bit.

1091 — Concert Tickets

Tickets have prices h_i. Customers arrive in order with maximum prices t_i. Give each customer the most expensive remaining ticket with price $\le t_i$; each ticket is sold at most once. **In:** `n m`; prices `h_i`; customer limits `t_i`

Out: price paid by each customer, or `-1`

Constraints: $1 \le n, m \le 2 \cdot 10^5$, $1 \le h_i, t_i \le 10^9$.

1141 — Playlist

Given a sequence of song IDs, find the longest contiguous segment in which every song ID is distinct. **In:** `n`; then `k_1 ... k_n`

Out: maximum length

Constraints: $1 \le n \le 2 \cdot 10^5$, $1 \le k_i \le 10^9$.

1163 — Traffic Lights

A street spans positions 0 through x. Insert n traffic lights one by one at distinct positions. After every insertion, find the longest interval containing no traffic light in its interior. **In:** `x n`; then positions `p_1 ... p_n`

Out: n longest passage lengths, after each insertion

Constraints: $1 \le x \le 10^9$, $n \le 2 \cdot 10^5$, $0 < p_i < x$.

1164 — Room Allocation

Customer i stays from day a_i through b_i. A room can be reused only if the previous customer's departure day is *earlier* than the next arrival day. Minimize rooms and give an allocation. **In:** `n`; then `a b` for each customer

Out: minimum room count k; then each customer's room in input order

Constraints: $1 \le n \le 2 \cdot 10^5$, $1 \le a \le b \le 10^9$.

1619 — Restaurant Customers

Given arrival and leaving times of n customers, find the maximum number simultaneously in the restaurant. All arrival/leaving times are distinct. **In:** `n`; then `a b` for each customer

Out: maximum simultaneous customers

Constraints: $1 \le n \le 2 \cdot 10^5$, $1 \le a < b \le 10^9$.

1620 — Factory Machines

Machine i needs k_i seconds per product and machines work simultaneously. Find the minimum time needed to produce at least t products in total. **In:** `n t`; then `k_1 ... k_n`

Out: minimum required time

Constraints: $n \le 2 \cdot 10^5$, $t \le 10^9$, $k_i \le 10^9$. Use 64-bit.

1629 — Movie Festival

Given start/end times of n movies, choose a maximum-size set that one person can watch completely without overlaps. **In:** `n`; then `a b` for each movie

Out: maximum number of movies

Constraints: $1 \le n \le 2 \cdot 10^5$, $1 \le a < b \le 10^9$.

1630 — Tasks and Deadlines

Process every task exactly once, sequentially from time 0. Task i has duration a_i and deadline d_i; if it finishes at f_i, its reward is $d_i - f_i$. Maximize total reward. **In:** `n`; then `a d` for each task

Out: maximum total reward

Constraints: $n \le 2 \cdot 10^5$, $1 \le a, d \le 10^6$. Use 64-bit.

1631 — Reading Books

Two readers must each read all n books from beginning to end. Book i takes t_i time, and the two readers cannot read the same book simultaneously. Minimize the total finishing time. **In:** `n`; then `t_1 ... t_n`

Out: minimum total time

Constraints: $1 \le n \le 2 \cdot 10^5$, $1 \le t_i \le 10^9$. Use 64-bit.

1632 — Movie Festival II

There are k club members and n movies with start/end times. Each member may watch only non-overlapping movies. Maximize the total number watched across all members. **In:** `n k`; then `a b` for each movie

Out: maximum total number of movies

Constraints: $1 \le k \le n \le 2 \cdot 10^5$, $1 \le a < b \le 10^9$.

1640 — Sum of Two Values

Find two array values at distinct positions whose sum is exactly x. **In:** `n x`; then `a_1 ... a_n`

Out: two 1-indexed positions, or `IMPOSSIBLE`

Constraints: $n \le 2 \cdot 10^5$, $1 \le x, a_i \le 10^9$.

1641 — Sum of Three Values

Find three array values at distinct positions whose sum is exactly x. **In:** `n x`; then `a_1 ... a_n`

Out: three 1-indexed positions, or `IMPOSSIBLE`

Constraints: $1 \le n \le 5000$, $1 \le x, a_i \le 10^9$.

1642 — Sum of Four Values

Find four array values at distinct positions whose sum is exactly x. **In:** `n x`; then `a_1 ... a_n`

Out: four 1-indexed positions, or `IMPOSSIBLE`

Constraints: $1 \le n \le 1000$, $1 \le x, a_i \le 10^9$.

1643 — Maximum Subarray Sum
Find the maximum sum of a contiguous, nonempty subarray. **In:** `n`; then `x_1 ... x_n`
Out: maximum subarray sum
Constraints: $1 \le n \le 2 \cdot 10^5$, $-10^9 \le x_i \le 10^9$. Use 64-bit.

1644 — Maximum Subarray Sum II
Find the maximum sum of a contiguous subarray whose length is between a and b, inclusive. **In:** `n a b`; then `x_1 ... x_n`
Out: maximum valid subarray sum
Constraints: $n \le 2 \cdot 10^5$, $1 \le a \le b \le n$, $|x_i| \le 10^9$. Use 64-bit.

1645 — Nearest Smaller Values
For every array position i, find the nearest position $j < i$ with $x_j < x_i$. **In:** `n`; then `x_1 ... x_n`
Out: n positions; print 0 when no such j exists
Constraints: $1 \le n \le 2 \cdot 10^5$, $1 \le x_i \le 10^9$.

1660 — Subarray Sums I
Given an array of *positive* integers, count contiguous subarrays whose sum is exactly x. **In:** `n x`; then `a_1 ... a_n`
Out: number of matching subarrays
Constraints: $n \le 2 \cdot 10^5$, $1 \le x, a_i \le 10^9$. Use 64-bit count.

1661 — Subarray Sums II
Given an integer array, count contiguous subarrays whose sum is exactly x. **In:** `n x`; then `a_1 ... a_n`
Out: number of matching subarrays
Constraints: $n \le 2 \cdot 10^5$, $-10^9 \le x, a_i \le 10^9$. Use 64-bit.

1662 — Subarray Divisibility
Count contiguous subarrays whose sum is divisible by n. **In:** `n`; then `a_1 ... a_n`
Out: number of valid subarrays
Constraints: $1 \le n \le 2 \cdot 10^5$, $-10^9 \le a_i \le 10^9$. Use 64-bit.

2162 — Josephus Problem I
Children $1, 2, \ldots, n$ stand in a circle. Repeatedly remove every other remaining child until none remain.
In: `n`
Out: the complete removal order
Constraints: $1 \le n \le 2 \cdot 10^5$.

2163 — Josephus Problem II
Children $1, 2, \ldots, n$ stand in a circle. Repeatedly skip k remaining children and remove the next child. **In:** `n k`
Out: the complete removal order
Constraints: $1 \le n \le 2 \cdot 10^5$, $0 \le k \le 10^9$.

2168 — Nested Ranges Check
For each distinct range $[x, y]$, determine (1) whether it contains some other input range, and (2) whether some other input range contains it. $[a, b]$ contains $[c, d]$ iff $a \le c$ and $d \le b$. **In:** `n`; then `x y` for each range
Out: two n-entry 0/1 lines, both in input order: contains; contained-by
Constraints: $n \le 2 \cdot 10^5$, $1 \le x < y \le 10^9$; ranges distinct.

2169 — Nested Ranges Count
For each distinct range $[x, y]$, count (1) how many other ranges it contains and (2) how many other ranges contain it, using inclusive containment. **In:** `n`; then `x y` for each range
Out: two n-entry lines in input order: contained count; containing count
Constraints: $n \le 2 \cdot 10^5$, $1 \le x < y \le 10^9$; ranges distinct.

2183 — Missing Coin Sum
Given positive coin values, each usable at most once, find the smallest positive integer sum that cannot be formed by a subset. **In:** `n`; then `x_1 ... x_n`
Out: smallest unformable positive sum
Constraints: $1 \le n \le 2 \cdot 10^5$, $1 \le x_i \le 10^9$. Use 64-bit.

2216 — Collecting Numbers
The array is a permutation of $1, \ldots, n$. In each round scan left-to-right, collecting as many next required numbers as possible while collecting $1, 2, \ldots, n$ in order. Find the number of rounds. **In:** `n`; then permutation `x_1 ... x_n`
Out: number of rounds
Constraints: $1 \le n \le 2 \cdot 10^5$.

2217 — Collecting Numbers II
Same collecting process as above. Perform m operations, each swapping the values at two positions a, b. Report the required number of rounds after every swap.
In: `n m`; permutation; then m swaps `a b`
Out: m answers, one after each swap
Constraints: $1 \le n, m \le 2 \cdot 10^5$, $1 \le a, b \le n$.

2428 — Distinct Values Subarrays II
Count contiguous subarrays containing at most k distinct values. **In:** `n k`; then `x_1 ... x_n`
Out: number of valid subarrays
Constraints: $1 \le k \le n \le 2 \cdot 10^5$, $1 \le x_i \le 10^9$. Use 64-bit.

3420 — Distinct Values Subarrays
Count contiguous subarrays in which every element is distinct. **In:** `n`; then `x_1 ... x_n`
Out: number of all-distinct subarrays
Constraints: $1 \le n \le 2 \cdot 10^5$, $1 \le x_i \le 10^9$. Use 64-bit.

3421 — Distinct Values Subsequences
Count nonempty subsequences in which every chosen element is distinct. Subsequences are determined by positions and may have gaps. Print the count modulo $10^9 + 7$. **In:** `n`; then `x_1 ... x_n`
Out: number of all-distinct subsequences mod $10^9 + 7$
Constraints: $1 \le n \le 2 \cdot 10^5$, $1 \le x_i \le 10^9$.

CSES Dynamic Programming — Problem Reference

All listed tasks: 1 s, 512 MB. Examples omitted for compactness.

1093 — Two Sets II
Count the ways to partition $\{1, 2, \ldots, n\}$ into two sets having equal sum. The two sets are unordered: swapping the two sides is the same partition. **In:** `n`
Out: number of partitions modulo $10^9 + 7$
Constraints: $1 \le n \le 500$.

1097 — Removal Game
Two players alternately remove the first or last number of a list and add it to their score. Both play optimally. Find the maximum final score the first player can guarantee. **In:** `n`; then `x_1 ... x_n`
Out: maximum score of the first player
Constraints: $1 \le n \le 5000$, $-10^9 \le x_i \le 10^9$. Use 64-bit.

1140 — Projects
Project i occupies every day from a_i through b_i and gives reward p_i. Choose non-overlapping projects, with at most one attended on any day, maximizing total reward. **In:** `n`; then `a_i b_i p_i`
Out: maximum total reward
Constraints: $n \le 2 \cdot 10^5$, $1 \le a_i \le b_i \le 10^9$, $p_i \le 10^9$. Use 64-bit.

1145 — Increasing Subsequence
Given an integer array, find the length of its longest *strictly increasing* subsequence. A subsequence preserves order but may skip elements. **In:** `n`; then `x_1 ... x_n`
Out: LIS length
Constraints: $1 \le n \le 2 \cdot 10^5$, $1 \le x_i \le 10^9$.

1158 — Book Shop
There are n books; book i costs h_i and has s_i pages. Buy each book at most once, spending at most x, and maximize total pages. **In:** `n x`; prices `h_i`; pages `s_i`
Out: maximum number of pages
Constraints: $n \le 1000$, $x \le 10^5$, $1 \le h_i, s_i \le 1000$.

1633 — Dice Combinations
Count ordered sequences of dice throws from $\{1, 2, \ldots, 6\}$ whose sum is exactly n. **In:** `n`
Out: number of sequences modulo $10^9 + 7$
Constraints: $1 \le n \le 10^6$.

1634 — Minimizing Coins
Given distinct positive coin values, each usable any number of times, form sum x using the minimum number of coins. **In:** `n x`; then `c_1 ... c_n`
Out: minimum coin count, or `-1` if impossible
Constraints: $n \le 100$, $x \le 10^6$, $1 \le c_i \le 10^6$.

1635 — Coin Combinations I
Given distinct positive coin values, each reusable, count *ordered* sequences of coins whose values sum to x. Different orders count separately. **In:** `n x`; then `c_1 ... c_n`
Out: number of ordered combinations modulo $10^9 + 7$
Constraints: $n \le 100$, $x \le 10^6$, $1 \le c_i \le 10^6$.

1636 — Coin Combinations II
Given distinct positive coin values, each reusable, count combinations whose values sum to x when order does *not* matter. **In:** `n x`; then `c_1 ... c_n`
Out: number of combinations modulo $10^9 + 7$
Constraints: $n \le 100$, $x \le 10^6$, $1 \le c_i \le 10^6$.

1637 — Removing Digits
Starting from n, each step subtracts one decimal digit currently appearing in the number. Find the minimum steps needed to reach 0. **In:** `n`
Out: minimum number of steps
Constraints: $1 \le n \le 10^6$.

1638 — Grid Paths I
An $n \times n$ grid contains empty cells `.` and traps `*`. Move from the upper-left to the lower-right using only right/down moves and never enter a trap. **In:** `n`; then n grid rows
Out: number of valid paths modulo $10^9 + 7$
Constraints: $1 \le n \le 1000$.

1639 — Edit Distance
Find the minimum number of single-character insertions, deletions, and replacements needed to transform one uppercase string into the other. **In:** first string; second string
Out: edit distance
Constraints: $1 \le n, m \le 5000$.

1653 — Elevator Rides
There are n people with weights w_i and an elevator capacity x. Every person must reach the top; each ride may carry any subset whose total weight is at most x. Minimize rides. **In:** `n x`; then `w_1 ... w_n`
Out: minimum number of rides
Constraints: $1 \le n \le 20$, $1 \le w_i \le x \le 10^9$.

1744 — Rectangle Cutting
Cut an $a \times b$ integer-sided rectangle into squares. Each move cuts one current rectangle into two rectangles along an integer coordinate. Minimize the number of cuts. **In:** `a b`
Out: minimum number of cuts
Constraints: $1 \le a, b \le 500$.

1745 — Money Sums
Given n positive-value coins, each usable at most once, find every positive sum obtainable from a subset. **In:** `n`; then `x_1 ... x_n`
Out: number k of distinct positive sums; then all sums in increasing order
Constraints: $1 \le n \le 100$, $1 \le x_i \le 1000$.

1746 — Array Description
Count arrays of length n with values in $[1, m]$ such that adjacent values differ by at most 1. Input positions may be fixed; value 0 means unknown. **In:** `n m`; then `x_1 ... x_n`
Out: number of matching arrays modulo $10^9 + 7$
Constraints: $n \le 10^5$, $m \le 100$, $0 \le x_i \le m$.

1748 — Increasing Subsequence II
Count all nonempty *strictly increasing* subsequences of an integer array. Subsequences using different positions are distinct even if their values match. **In:** `n`; then `x_1 ... x_n`
Out: number of increasing subsequences modulo $10^9 + 7$
Constraints: $1 \le n \le 2 \cdot 10^5$, $1 \le x_i \le 10^9$.

2181 — Counting Tilings
Count ways to tile an $n \times m$ rectangle completely using 1×2 and 2×1 dominoes. **In:** `n m`
Out: number of tilings modulo $10^9 + 7$
Constraints: $1 \le n \le 10$, $1 \le m \le 1000$.

2220 — Counting Numbers
Count integers in the inclusive range $[a, b]$ whose decimal representation has no two equal adjacent digits. **In:** `a b`
Out: number of valid integers
Constraints: $0 \le a \le b \le 10^{18}$. Use 64-bit.

2413 — Counting Towers
For each query n, count different towers of width 2 and height n built from integer-sided blocks. Distinct-looking mirrored or rotated constructions count separately. **In:** `t`; then one `n` per test
Out: answer for each test modulo $10^9 + 7$
Constraints: $1 \le t \le 100$, $1 \le n \le 10^6$.

3314 — Mountain Range
Mountains stand in a row. You may glide from mountain a to b when a is taller than b and taller than every mountain strictly between them. Start anywhere. Maximize mountains visited. **In:** `n`; then `h_1 ... h_n`
Out: maximum number of visited mountains
Constraints: $1 \le n \le 2 \cdot 10^5$, $1 \le h_i \le 10^9$.

3359 — Minimal Grid Path
An $n \times n$ grid contains uppercase letters. Move from the upper-left to the lower-right using only right/down moves; concatenate every visited cell. Find the lexicographically smallest possible string. **In:** `n`; then n rows of n letters
Out: lexicographically minimum path string
Constraints: $1 \le n \le 3000$.

3403 — Longest Common Subsequence
Given two integer arrays, find a longest sequence obtainable as a subsequence of both. A subsequence preserves order but may skip elements. **In:** `n m`; array `a`; array `b`
Out: LCS length; then any one longest common subsequence
Constraints: $1 \le n, m \le 1000$, $1 \le a_i, b_i \le 10^9$.

CSES Graph Algorithms — Problem Reference

All listed tasks: 1 s; 512 MB unless noted. Examples omitted for compactness.

1160 — Planets Queries II

Each planet has exactly one outgoing teleporter t_i. For each query (a, b), find the minimum number of teleportations needed to reach b from a, or determine that it is impossible. **In:** `n q`; destinations `t_1 ... t_n`; then `a b` queries
Out: minimum jumps for each query, or `-1`
Constraints: $1 \le n, q \le 2 \cdot 10^5$, $1 \le t_i, a, b \le n$.

1192 — Counting Rooms

An $n \times m$ map contains floor `.` and walls `#`. Moving in four directions through floor cells, count the connected floor components (rooms).
In: `n m`; then n grid rows
Out: number of rooms
Constraints: $1 \le n, m \le 1000$.

1193 — Labyrinth

In a grid with walls `#`, floor `.`, one start `A`, and one target `B`, find a shortest four-direction path from A to B. **In:** `n m`; then n grid rows
Out: `NO`, or `YES`, path length, and any shortest path using `L/R/U/D`
Constraints: $1 \le n, m \le 1000$.

1194 — Monsters

You and all monsters move simultaneously one grid step per turn. Escape from `A` to any boundary cell without ever occupying a cell at the same time as a monster; the route must be safe even if monsters know it. **In:** `n m`; grid with `.`, `#`, one `A`, and zero or more `M`
Out: `NO`, or `YES`, a valid path length and `D/U/L/R` path
Constraints: $1 \le n, m \le 1000$; printed path length $\le nm$.

1195 — Flight Discount

Directed flights have positive prices. Travel from city 1 to city n and use one coupon on a single edge, changing price c to $\lfloor c/2 \rfloor$. Minimize total cost. **In:** `n m`; then directed edges `a b c`
Out: minimum route price
Constraints: $n \le 10^5$, $m \le 2 \cdot 10^5$, $c \le 10^9$; a route exists. Use 64-bit.

1196 — Flight Routes

Find the k cheapest directed routes from city 1 to city n. Routes may revisit vertices, equal-cost routes count separately, and at least k routes exist. **In:** `n m k`; then directed edges `a b c`
Out: k route costs in nondecreasing order
Constraints: $n \le 10^5$, $m \le 2 \cdot 10^5$, $c \le 10^9$, $1 \le k \le 10$. Use 64-bit.

1197 — Cycle Finding

Given a directed weighted graph, determine whether it contains a cycle whose total edge weight is negative, and reconstruct one if it exists. **In:** `n m`; then directed edges `a b c`
Out: `NO`, or `YES` followed by the cycle vertices in order (start repeated at end)
Constraints: $n \le 2500$, $m \le 5000$, $|c| \le 10^9$. Use 64-bit.

1202 — Investigation

For directed positive-cost flights from 1 to n, determine: minimum route price, number of minimum-price routes modulo $10^9 + 7$, minimum flights among them, and maximum flights among them. **In:** `n m`; then directed edges `a b c`
Out: four integers: distance, count, min edges, max edges
Constraints: $n \le 10^5$, $m \le 2 \cdot 10^5$, $c \le 10^9$; a route exists. Use 64-bit.

1666 — Building Roads

Given an undirected graph, add the minimum number of roads so that every pair of cities becomes connected. Output any set of roads achieving this. **In:** `n m`; then undirected edges `a b`
Out: number k of added roads; then k road endpoints
Constraints: $n \le 10^5$, $m \le 2 \cdot 10^5$; no loops or duplicate roads.

1667 — Message Route

In an undirected unweighted network, find a route from computer 1 to n using the minimum number of computers (equivalently, shortest number of edges) and reconstruct it. **In:** `n m`; then undirected edges `a b`
Out: `IMPOSSIBLE`, or route vertex count and any shortest route
Constraints: $2 \le n \le 10^5$, $m \le 2 \cdot 10^5$.

1668 — Building Teams

Split the vertices of an undirected friendship graph into two teams so every edge has endpoints in different teams. **In:** `n m`; then undirected edges `a b`
Out: a label `1` or `2` for each vertex, or `IMPOSSIBLE`
Constraints: $n \le 10^5$, $m \le 2 \cdot 10^5$.

1669 — Round Trip

Find any simple cycle in an undirected graph: the route must start/end at the same city and all intermediate cities must be distinct. **In:** `n m`; then undirected edges `a b`
Out: `IMPOSSIBLE`, or cycle vertex count and vertices in order
Constraints: $n \le 10^5$, $m \le 2 \cdot 10^5$; no loops or parallel edges.

1671 — Shortest Routes I

In a directed graph with positive edge lengths, find the shortest distance from city 1 to every city. All cities are reachable from 1. **In:** `n m`; then directed edges `a b c`
Out: n distances from city 1
Constraints: $n \le 10^5$, $m \le 2 \cdot 10^5$, $c \le 10^9$. Use 64-bit.

1672 — Shortest Routes II

Given an undirected weighted graph and many queries, answer the shortest distance between each queried pair of cities; pairs may be disconnected. **In:** `n m q`; m undirected edges `a b c`; then q queries `a b`
Out: distance per query, or `-1` if unreachable
Constraints: $n \le 500$, $m \le n^2$, $q \le 10^5$, $c \le 10^9$. Use 64-bit.

1673 — High Score

Directed edges may add positive or negative score. Starting at 1, reach n with maximum total score; edges may be reused. If the score can be made arbitrarily large on some valid $1 \to n$ route, print -1. **In:** `n m`; then directed edges `a b x`
Out: maximum score, or `-1` if unbounded
Constraints: $n \le 2500$, $m \le 5000$, $|x| \le 10^9$; n is reachable from 1. Use 64-bit.

1675 — Road Reparation

Undirected roads have repair costs. Choose roads of minimum total cost so the repaired network is connected; if impossible, report it.
In: `n m`; then undirected roads `a b c`
Out: minimum total repair cost, or `IMPOSSIBLE`
Constraints: $n \le 10^5$, $m \le 2 \cdot 10^5$, $c \le 10^9$. **Memory 128 MB**. Use 64-bit.

1676 — Road Construction

Start with n isolated cities and add m undirected roads one by one. After each addition, report the current number of connected components and the size of the largest component.
In: `n m`; then roads `a b` in insertion order
Out: after each road: component count and largest component size
Constraints: $n \le 10^5$, $m \le 2 \cdot 10^5$.

1678 — Round Trip II

Find any directed cycle: start/end at the same city and all intermediate cities must be distinct.
In: `n m`; then directed edges `a b`
Out: `IMPOSSIBLE`, or cycle vertex count and vertices in order
Constraints: $n \le 10^5$, $m \le 2 \cdot 10^5$.

1679 — Course Schedule
Given prerequisites $a \to b$ meaning course a must come before b, output any ordering containing every course that satisfies all requirements. **In:** `n m`; then prerequisite edges `a b`
Out: a valid course order, or `IMPOSSIBLE`
Constraints: $n \le 10^5$, $m \le 2 \cdot 10^5$.

1680 — Longest Flight Route
The directed flight graph is a DAG. Find a route from city 1 to city n that visits the maximum possible number of cities, and reconstruct it. **In:** `n m`; then directed edges `a b`
Out: `IMPOSSIBLE`, or maximum city count and any maximizing route
Constraints: $2 \le n \le 10^5$, $m \le 2 \cdot 10^5$; graph is acyclic.

1681 — Game Routes
A directed acyclic graph has teleporters between levels. Count the number of directed routes from level 1 to level n. **In:** `n m`; then directed edges `a b`
Out: number of routes modulo $10^9 + 7$
Constraints: $n \le 10^5$, $m \le 2 \cdot 10^5$; graph is acyclic.

1682 — Flight Routes Check
Determine whether a directed graph is strongly connected, i.e. every city can reach every other city. **In:** `n m`; then directed edges `a b`
Out: `YES`, or `NO` and any pair `a b` such that a cannot reach b
Constraints: $n \le 10^5$, $m \le 2 \cdot 10^5$.

1683 — Planets and Kingdoms
Partition a directed graph into strongly connected components: two planets share a kingdom exactly when each can reach the other.
In: `n m`; then directed edges `a b`
Out: number k of SCCs; then one kingdom label $1..k$ for each planet
Constraints: $n \le 10^5$, $m \le 2 \cdot 10^5$.

1684 — Giant Pizza
There are m Boolean toppings and n clauses. Each clause contains two literals `+ x` or `- x`; choose topping truth values so at least one literal in every clause is true. **In:** `n m`; then n pairs of signed topping literals
Out: one `+/-` choice for each topping, or `IMPOSSIBLE`
Constraints: $1 \le n, m \le 10^5$, topping indices $1..m$.

1686 — Coin Collector
Each room has k_i coins and directed tunnels connect rooms. You may start and end anywhere and move along tunnels; maximize the total coins obtainable from visited rooms. **In:** `n m`; coin values `k_1 ... k_n`; then directed edges `a b`
Out: maximum collectible coins
Constraints: $n \le 10^5$, $m \le 2 \cdot 10^5$, $k_i \le 10^9$. Use 64-bit.

1689 — Knight's Tour
Given a starting square on an 8×8 board, construct any knight's tour that visits every square exactly once. **In:** `x y`, the starting position
Out: an 8×8 grid whose entries 1..64 give the visit order
Constraints: $1 \le x, y \le 8$.

1690 — Hamiltonian Flights
Count directed routes from city 1 to city n that visit every city exactly once. **In:** `n m`; then directed edges `a b`
Out: number of Hamiltonian $1 \to n$ routes modulo $10^9 + 7$
Constraints: $2 \le n \le 20$, $1 \le m \le n^2$.

1691 — Mail Delivery
In an undirected graph, find a closed route starting and ending at crossing 1 that uses every street exactly once. **In:** `n m`; then undirected edges `a b`
Out: an Euler circuit as $m + 1$ crossings, or `IMPOSSIBLE`
Constraints: $2 \le n \le 10^5$, $m \le 2 \cdot 10^5$; no loops or parallel streets.

1692 — De Bruijn Sequence
Construct a minimum-length binary string containing every possible binary string of length n as a contiguous substring. **In:** `n`
Out: any minimum-length valid bit string
Constraints: $1 \le n \le 15$.

1693 — Teleporters Path
In a directed graph, start at level 1, finish at level n, and use every teleporter (edge) exactly once. **In:** `n m`; then distinct directed edges `a b`
Out: an Euler trail as $m + 1$ vertices, or `IMPOSSIBLE`
Constraints: $2 \le n \le 10^5$, $m \le 2 \cdot 10^5$.

1694 — Download Speed
A directed network edge $a \to b$ has capacity c. Find the maximum total data flow from server 1 to computer n. **In:** `n m`; then directed capacities `a b c`
Out: maximum flow value
Constraints: $n \le 500$, $m \le 1000$, $c \le 10^9$. Use 64-bit.

1695 — Police Chase
An undirected street graph connects bank 1 to harbor n. Find the minimum number of streets whose removal disconnects them, and output one such minimum cut. **In:** `n m`; then undirected streets `a b`
Out: cut size k; then the k streets to close
Constraints: $2 \le n \le 500$, $m \le 1000$; at most one street per pair.

1696 — School Dance
Given a bipartite graph of n boys, m girls, and k acceptable pairs, find a maximum matching and output the matched pairs. **In:** `n m k`; then acceptable pairs `boy girl`
Out: matching size r; then r matched pairs
Constraints: $n, m \le 500$, $k \le 1000$.

1711 — Distinct Routes
Directed teleporters may each be used on at most one day. Every day starts at room 1 and ends at room n. Maximize the number of edge-disjoint $1 \to n$ routes and output them.
In: `n m`; then distinct directed edges `a b`
Out: maximum route count k; then for each route its vertex count and vertices
Constraints: $2 \le n \le 500$, $m \le 1000$.

1750 — Planets Queries I
Each planet has one outgoing teleporter t_i. For each query (x, k), determine the planet reached after exactly k teleportations from x. **In:** `n q`; destinations `t_1 ... t_n`; then queries `x k`
Out: destination planet for each query
Constraints: $n, q \le 2 \cdot 10^5$, $0 \le k \le 10^9$.

1751 — Planets Cycles
Each planet has one outgoing teleporter. Starting from each planet, keep teleporting until the next planet has already been visited in that walk. Report the number of teleportations performed. **In:** `n`; destinations `t_1 ... t_n`
Out: n answers, one for each starting planet
Constraints: $1 \le n \le 2 \cdot 10^5$, $1 \le t_i \le n$.

CSES — Range Queries Problem Reference

All tasks: 1 s, 512 MB. Ranges and array indices are 1-based.

1143 — Hotel Queries

n hotels have free-room counts h_i. For each arriving group requiring r, assign the *first* hotel with at least r rooms, then decrease that hotel's free rooms by r; output 0 if none exists. **In:** `n m`; `h_1 ... h_n`; `r_1 ... r_m`
Out: assigned hotel index for every group, or 0
Constraints: $n, m \le 2 \cdot 10^5$; $h_i, r_i \le 10^9$.

1144 — Salary Queries

Maintain salaries p_i under point assignments. Query how many employees have salary in the inclusive value interval $[a, b]$. **In:** `n q`; salaries; then `! k x` or `? a b`
Out: count for each `?` query
Constraints: $n, q \le 2 \cdot 10^5$; salaries/query values $\le 10^9$.

1190 — Subarray Sum Queries

Maintain an integer array under point assignments. After *every* update, report the maximum subarray sum of the entire array; the empty subarray with sum 0 is allowed. **In:** `n m`; array; then m updates `k x`
Out: maximum subarray sum after each update
Constraints: $n, m \le 2 \cdot 10^5$; $|x_i|, |x| \le 10^9$. Use 64-bit.

1646 — Static Range Sum Queries

Given a fixed array, answer the sum of each inclusive range $[a, b]$. **In:** `n q`; array; then q lines `a b`
Out: $\sum_{i=a}^{b} x_i$ for each query
Constraints: $n, q \le 2 \cdot 10^5$; $x_i \le 10^9$. Use 64-bit.

1647 — Static Range Minimum Queries

Given a fixed array, answer the minimum value in each inclusive range $[a, b]$. **In:** `n q`; array; then q lines `a b`
Out: $\min(x_a, \dots, x_b)$ for each query
Constraints: $n, q \le 2 \cdot 10^5$; $x_i \le 10^9$.

1648 — Dynamic Range Sum Queries

Maintain an array with point assignment and range-sum queries. **In:** `n q`; array; queries `1 k u` (set $x_k = u$) or `2 a b` (sum)
Out: range sum for each type-2 query
Constraints: $n, q \le 2 \cdot 10^5$; $x_i, u \le 10^9$. Use 64-bit.

1649 — Dynamic Range Minimum Queries

Maintain an array with point assignment and range-minimum queries. **In:** `n q`; array; queries `1 k u` (set) or `2 a b` (minimum)
Out: range minimum for each type-2 query
Constraints: $n, q \le 2 \cdot 10^5$; $x_i, u \le 10^9$.

1650 — Range Xor Queries

Given a fixed array, answer the bitwise xor of values in each inclusive range $[a, b]$. **In:** `n q`; array; then q lines `a b`
Out: $x_a \oplus \cdots \oplus x_b$ for each query
Constraints: $n, q \le 2 \cdot 10^5$; $x_i \le 10^9$.

1651 — Range Update Queries

Maintain an array under range additions. Type 1 adds u to every x_i for $i \in [a, b]$; type 2 asks the current value at one position k. **In:** `n q`; array; `1 a b u` or `2 k`
Out: x_k for each type-2 query
Constraints: $n, q \le 2 \cdot 10^5$; $x_i, u \le 10^9$. Use 64-bit.

1652 — Forest Queries

An $n \times n$ fixed grid contains trees `*` and empty cells `.`. For each rectangle with corners (y_1, x_1) and (y_2, x_2), count trees inside it, boundaries included. **In:** `n q`; n grid rows; then `y1 x1 y2 x2`
Out: tree count for each rectangle
Constraints: $n \le 1000$, $q \le 2 \cdot 10^5$.

1664 — Movie Festival Queries

n movies have intervals $[s, e]$. For each arrival/leaving interval $[a, b]$, find the maximum number of whole movies that can be watched: first movie may start at a, consecutive movies may touch ($e_1 \le s_2$), and last may end at b. **In:** `n q`; n movie lines `a b`; then q availability lines `a b`
Out: maximum watchable movies for each query
Constraints: $n, q \le 2 \cdot 10^5$; all times in $[1, 10^6]$, start $<$ end.

1734 — Distinct Values Queries

Given a fixed array, count the number of distinct values in each range $[a, b]$. **In:** `n q`; array; then q lines `a b`
Out: number of distinct values in each range
Constraints: $n, q \le 2 \cdot 10^5$; $x_i \le 10^9$.

1735 — Range Updates and Sums

Maintain an array with three operations: add x to all values in $[a, b]$; set all values in $[a, b]$ to x; or query the sum of $[a, b]$. **In:** `n q`; array; then `1 a b x`, `2 a b x`, or `3 a b`
Out: range sum for each type-3 query
Constraints: $n, q \le 2 \cdot 10^5$; initial/update values $\le 10^6$. Use 64-bit.

1736 — Polynomial Queries

Type 1 on $[a, b]$ adds $1, 2, \dots, b - a + 1$ respectively to $x_a, x_{a+1}, \dots, x_b$. Type 2 asks the sum of $[a, b]$. **In:** `n q`; array; then `1 a b` or `2 a b`
Out: range sum for each type-2 query
Constraints: $n, q \le 2 \cdot 10^5$; $x_i \le 10^6$. Use 64-bit.

1737 — Range Queries and Copies

Maintain a list of versioned arrays, initially containing one array. Operations: set position a in version k to x; query sum $[a, b]$ in version k; or append an independent copy of version k as a new version. **In:** `n q`; array; then `1 k a x`, `2 k a b`, or `3 k`
Out: range sum for each type-2 query
Constraints: $n, q \le 2 \cdot 10^5$; $t_i, x \le 10^9$. Use 64-bit.

1739 — Forest Queries II

Maintain an $n \times n$ tree grid. Type 1 toggles cell (y, x) between empty/tree; type 2 counts trees in an inclusive query rectangle. **In:** `n q`; grid; then `1 y x` or `2 y1 x1 y2 x2`
Out: tree count for each type-2 query
Constraints: $n \le 1000$, $q \le 2 \cdot 10^5$.

1749 — List Removals

Start with list $x_1, \dots, x_n$. Repeatedly remove and print the element currently at position p_i; positions are relative to the shrinking list. **In:** `n`; values `x_1 ... x_n`; removals `p_1 ... p_n`
Out: removed values in order
Constraints: $n \le 2 \cdot 10^5$; $x_i \le 10^9$; $1 \le p_i \le n - i + 1$.

2166 — Prefix Sum Queries

Maintain an array under point assignments. For a query range $[a, b]$, report the maximum prefix sum $\max(0, x_a, x_a + x_{a+1}, \dots, \sum_{i=a}^{b} x_i)$. **In:** `n q`; array; `1 k u` (set) or `2 a b` (query)
Out: maximum prefix sum for each type-2 query
Constraints: $n, q \le 2 \cdot 10^5$; $|x_i|, |u| \le 10^9$. Use 64-bit.

2184 — Missing Coin Sum Queries

Coins have positive values x_i. For each index range $[a, b]$, using each coin there at most once, find the smallest positive sum that cannot be formed. **In:** `n q`; coin values; then q lines `a b`
Out: smallest nonrepresentable positive sum for each query
Constraints: $n, q \le 2 \cdot 10^5$; $x_i \le 10^9$. Use 64-bit.

2206 — Pizzeria Queries

Building i sells pizza for p_i. Delivery from pizzeria a to customer at k costs $p_a + |a - k|$. Support point price changes and queries for the minimum delivered price at building k. **In:** `n q`; prices; then `1 k x` or `2 k`
Out: $\min_a(p_a + |a - k|)$ for each type-2 query
Constraints: $n, q \le 2 \cdot 10^5$; $p_i, x \le 10^9$. Use 64-bit.

2416 — Increasing Array Queries

For each subarray $x_a, \dots, x_b$, you may only increment elements. Find the minimum total number of $+1$ operations needed to make it nondecreasing ($x_i \ge x_{i-1}$). **In:** `n q`; array; then q lines `a b`
Out: minimum increments for each query
Constraints: $n, q \le 2 \cdot 10^5$; $x_i \le 10^9$. Use 64-bit.

3163 — Range Interval Queries

Given fixed $x_1, \dots, x_n$, each query (a, b, c, d) asks how many indices satisfy both $a \le i \le b$ and $c \le x_i \le d$. **In:** `n q`; array; then `a b c d`
Out: matching index count for each query
Constraints: $n, q \le 2 \cdot 10^5$; $x_i, c, d \le 10^9$.

3226 — Subarray Sum Queries II

Given a fixed integer array, for each range $[a, b]$ report the maximum subarray sum contained entirely inside that range. Empty subarrays with sum 0 are allowed. **In:** `n q`; array; then q lines `a b`
Out: maximum subarray sum in each query range
Constraints: $n, q \le 2 \cdot 10^5$; $|x_i| \le 10^9$. Use 64-bit.

3304 — Visible Buildings Queries

For each query $[a, b]$, pretend only those buildings exist and view them from the left. Count buildings whose height is *strictly greater* than every earlier height in the range. **In:** `n q`; heights; then q lines `a b`
Out: number of visible buildings per query
Constraints: $n \le 10^5$, $q \le 2 \cdot 10^5$; $h_i \le 10^9$.

3356 — Distinct Values Queries II

Maintain an array under point assignments. For each range query $[a, b]$, determine whether every value in the range is pairwise distinct. **In:** `n q`; array; then `1 k u` or `2 a b`
Out: `YES` if all values in $[a, b]$ are distinct, otherwise `NO`
Constraints: $n, q \le 2 \cdot 10^5$; $x_i, u \le 10^9$.

CSES — Tree Algorithms Reference

All tasks: 1 s, 512 MiB. Tree distances and path lengths are measured in edges.

1674 — Subordinates

Company hierarchy is rooted at employee 1. For every employee, count all descendants below them (not just direct reports). **In:** `n`; bosses of employees $2, \dots, n$

Out: n subordinate counts

Constraints: $n \le 2 \cdot 10^5$.

1130 — Tree Matching

Given an undirected tree, find the largest set of edges such that no vertex is incident to more than one chosen edge. **In:** `n`; then $n-1$ edges

Out: maximum matching size (number of edges)

Constraints: $n \le 2 \cdot 10^5$.

1131 — Tree Diameter

Find the maximum distance between any two vertices of an unweighted tree. **In:** `n`; then $n-1$ edges

Out: tree diameter in edges

Constraints: $n \le 2 \cdot 10^5$.

1132 — Tree Distances I

For every vertex v, find its eccentricity: the maximum distance from v to any other vertex. **In:** `n`; then $n-1$ edges

Out: n maximum distances, in vertex order

Constraints: $n \le 2 \cdot 10^5$.

1133 — Tree Distances II

For every vertex v, compute the sum of distances from v to all vertices of the tree. **In:** `n`; then $n-1$ edges

Out: n distance sums, in vertex order

Constraints: $n \le 2 \cdot 10^5$. Use 64-bit sums.

1687 — Company Queries I

The company hierarchy is rooted at employee 1. For each query (x, k), find the ancestor of employee x exactly k levels higher; report failure if it does not exist. **In:** `n q`; bosses of $2, \dots, n$; then queries `x k`

Out: the kth ancestor for each query, or `-1`

Constraints: $n, q \le 2 \cdot 10^5$, $1 \le k \le n$.

1688 — Company Queries II

For each pair of employees, find their lowest common ancestor in the hierarchy rooted at employee 1. **In:** `n q`; bosses of $2, \dots, n$; then queries `a b`

Out: LCA / lowest common boss for each query

Constraints: $n, q \le 2 \cdot 10^5$.

1135 — Distance Queries

For each query (a, b) on a fixed tree, report the number of edges on the unique path between a and b. **In:** `n q`; $n-1$ edges; then q pairs `a b`

Out: distance for each query

Constraints: $n, q \le 2 \cdot 10^5$.

1136 — Counting Paths

Given m queried tree paths, count for each vertex how many of those paths contain it. Both path endpoints count as contained vertices. **In:** `n m`; $n-1$ edges; then m endpoint pairs

Out: n path-containment counts

Constraints: $n, m \le 2 \cdot 10^5$.

1137 — Subtree Queries

Root the tree at 1; each vertex has a value. Support point assignment $v_s \leftarrow x$ and queries for the sum of values in the entire subtree of s (including s). **In:** `n q`; values; edges; queries `1 s x / 2 s`

Out: subtree sum for each type-2 query

Constraints: $n, q \le 2 \cdot 10^5$, values $\le 10^9$. Use 64-bit sums.

1138 — Path Queries

Root the tree at 1; each vertex has a value. Support point assignment $v_s \leftarrow x$ and queries for the sum on the inclusive path from root 1 to s. **In:** `n q`; values; edges; queries `1 s x / 2 s`

Out: root-to-s path sum for each type-2 query

Constraints: $n, q \le 2 \cdot 10^5$, values $\le 10^9$. Use 64-bit sums.

2134 — Path Queries II

Each tree vertex has a value. Support point assignment $v_s \leftarrow x$ and maximum-value queries on the inclusive path between arbitrary vertices a and b. **In:** `n q`; values; edges; queries `1 s x / 2 a b`

Out: path maximum for each type-2 query

Constraints: $n, q \le 2 \cdot 10^5$, values $\le 10^9$.

1139 — Distinct Colors

Root the tree at 1. For every vertex, count how many distinct color values occur among all vertices in its subtree, including itself. **In:** `n`; colors $c_1, \dots, c_n$; then $n-1$ edges

Out: n distinct-color counts

Constraints: $n \le 2 \cdot 10^5$, $c_i \le 10^9$.

2079 — Finding a Centroid

Find any vertex whose removal leaves every connected component with at most $\lfloor n/2 \rfloor$ vertices (equivalently, rooting there gives no child subtree larger than half the tree). **In:** `n`; then $n-1$ edges

Out: any centroid vertex

Constraints: $n \le 2 \cdot 10^5$.

2080 — Fixed-Length Paths I

Count distinct simple tree paths containing exactly k edges. A path is determined by its unordered pair of endpoints. **In:** `n k`; then $n-1$ edges

Out: number of length-k paths

Constraints: $1 \le k \le n \le 2 \cdot 10^5$. Use 64-bit count.

2081 — Fixed-Length Paths II

Count distinct simple tree paths whose length is between k_1 and k_2 edges, inclusive. A path is determined by its unordered pair of endpoints. **In:** `n k_1 k_2`; then $n-1$ edges

Out: number of paths with $k_1 \le$ length $\le k_2$

Constraints: $1 \le k_1 \le k_2 \le n \le 2 \cdot 10^5$. Use 64-bit count.

CSES – Mathematics Problem Reference

All tasks: 1 s, 512 MB. Arithmetic is modulo 10^9+7 only where explicitly stated.

1079 – Binomial Coefficients

Compute n values $\binom{a}{b}$ modulo 10^9+7, where $0 \le b \le a$. **In:** `n`; then n lines `a b`
Out: each $\binom{a}{b}$ mod 10^9+7
Constraints: $n \le 10^5$; $a \le 10^6$.

1081 – Common Divisors

Given n positive integers, find two array elements whose greatest common divisor is as large as possible. **In:** `n`; `x_1 ... x_n`
Out: maximum possible gcd
Constraints: $2 \le n \le 2\cdot 10^5$; $x_i \le 10^6$.

1082 – Sum of Divisors

Let $\sigma(i)$ be the sum of divisors of i. Compute $\sum_{i=1}^{n}\sigma(i)$ mod 10^9+7. **In:** `n`
Out: the sum modulo 10^9+7
Constraints: $1 \le n \le 10^{12}$.

1095 – Exponentiation

For each query, compute a^b mod 10^9+7. By definition here, $0^0=1$. **In:** `n`; then n lines `a b`
Out: one modular power per query
Constraints: $n \le 2\cdot 10^5$; $0 \le a,b \le 10^9$.

1096 – Throwing Dice

Count ordered sequences of dice throws, each in $\{1,\ldots,6\}$, whose total sum is exactly n. **In:** `n`
Out: number of ways modulo 10^9+7
Constraints: $1 \le n \le 10^{18}$.

1098 – Nim Game II

There are n heaps. A move chooses one nonempty heap and removes 1, 2, or 3 sticks. Last move wins; determine the winner under optimal play. **In:** `t`; for each test: `n`; heap sizes
Out: `first` or `second` for each test
Constraints: $t,n \le 2\cdot 10^5$; $x_i \le 10^9$; total $n \le 2\cdot 10^5$.

1099 – Stair Game

Stairs $1\ldots n$ contain balls. A move chooses $k \ne 1$ with balls and moves any positive number of balls from stair k to $k-1$. Last move wins; with no legal move initially, second wins. **In:** `t`; for each test: `n`; `p_1 ... p_n`
Out: `first` or `second`
Constraints: $t,n \le 2\cdot 10^5$; $0 \le p_i \le 10^9$; total $n \le 2\cdot 10^5$.

1712 – Exponentiation II

For each query, compute a^{b^c} mod 10^9+7. By definition here, $0^0=1$. **In:** `n`; then n lines `a b c`
Out: one answer per query
Constraints: $n \le 10^5$; $0 \le a,b,c \le 10^9$.

1713 – Counting Divisors

For each given integer x, report the number of positive divisors of x. **In:** `n`; then n values x
Out: divisor count for each x
Constraints: $n \le 10^5$; $x \le 10^6$.

1715 – Creating Strings II

Given a lowercase string, count the number of distinct strings obtainable by permuting all of its characters. **In:** one lowercase string of length n
Out: count modulo 10^9+7
Constraints: $1 \le n \le 10^6$.

1716 – Distributing Apples

Distribute m identical apples among n children; each child may receive any nonnegative number. Count all distributions. **In:** `n m`
Out: number of distributions modulo 10^9+7
Constraints: $1 \le n,m \le 10^6$.

1717 – Christmas Party

n children each brought one gift. Count assignments of gifts to children such that nobody receives their own gift. **In:** `n`
Out: number of valid distributions modulo 10^9+7
Constraints: $1 \le n \le 10^6$.

1722 – Fibonacci Numbers

With $F_0=0$, $F_1=1$, $F_n=F_{n-2}+F_{n-1}$, compute F_n. **In:** `n`
Out: F_n mod 10^9+7
Constraints: $0 \le n \le 10^{18}$.

1723 – Graph Paths I

Given a directed graph, count paths from node 1 to node n that use exactly k edges. **In:** `n m k`; then m directed edges `a b`
Out: number of length-k paths modulo 10^9+7
Constraints: $n \le 100$; $m \le n(n-1)$; $k \le 10^9$.

1724 – Graph Paths II

Given a directed weighted graph, find the minimum total weight of a path from node 1 to node n using exactly k edges. **In:** `n m k`; then edges `a b c`
Out: minimum length, or -1 if no such path exists
Constraints: $n \le 100$; $m \le n(n-1)$; $k \le 10^9$; $c \le 10^9$.

1725 – Dice Probability

Throw a fair six-sided die n times. Find the probability that the total sum lies in $[a,b]$. **In:** `n a b`
Out: probability rounded to 6 decimals
Constraints: $n \le 100$; $1 \le a \le b \le 6n$.

1726 – Moving Robots

Initially every square of an 8×8 board has one robot. Each robot independently makes k random legal orthogonal moves, choosing uniformly among directions that stay on the board. Find the expected number of empty squares afterward. **In:** `k`
Out: expected empty squares, rounded to 6 decimals
Constraints: $1 \le k \le 100$.

1727 – Candy Lottery

n children independently receive a uniformly random integer number of candies from 1 to k. Find the expected maximum received by any child. **In:** `n k`
Out: expected maximum, rounded to 6 decimals
Constraints: $1 \le n,k \le 100$.

1728 – Inversion Probability

For each position i, choose x_i independently and uniformly from $1\ldots r_i$. Find the expected number of inversions (a,b) with $a<b$ and $x_a > x_b$. **In:** `n`; `r_1 ... r_n`
Out: expected inversions, rounded to 6 decimals
Constraints: $n \le 100$; $r_i \le 100$.

1729 – Stick Game

A heap has s sticks. Allowed removals are the distinct values in P, and $1 \in P$. For every initial size $s = 1, \dots, n$, classify the position as winning or losing under optimal play; last move wins. **In:** `n k`; `p_1 ... p_k`
Out: length-n string of `W/L`
Constraints: $n \le 10^6$; $k \le 100$; $p_i \le n$.

1730 – Nim Game I

There are n heaps. A move chooses one nonempty heap and removes any positive number of sticks. Last move wins; determine the winner under optimal play. **In:** `t`; for each test: `n`; heap sizes
Out: `first` or `second`
Constraints: $t, n \le 2 \cdot 10^5$; $x_i \le 10^9$; total $n \le 2 \cdot 10^5$.

2064 – Bracket Sequences I

Count valid parenthesis sequences of total length n. **In:** `n`
Out: number of sequences modulo $10^9 + 7$
Constraints: $1 \le n \le 10^6$.

2164 – Josephus Queries

Children $1 \dots n$ stand in a circle; repeatedly remove every second remaining child until none remain. For each query (n, k), report the k-th child removed. **In:** `q`; then q lines `n k`
Out: k-th removed child for each query
Constraints: $q \le 10^5$; $1 \le k \le n \le 10^9$.

2182 – Divisor Analysis

A positive integer is given by its prime factorization $\prod x_i^{k_i}$, with distinct primes x_i. Compute the number, sum, and product of all positive divisors. **In:** `n`; then n lines `x k`
Out: divisor count, divisor sum, divisor product; all modulo $10^9 + 7$
Constraints: $n \le 10^5$; $x \le 10^6$; $k \le 10^9$.

2185 – Prime Multiples

Given k distinct primes a_i and integer n, count integers in $1 \dots n$ divisible by at least one a_i. **In:** `n k`; `a_1 ... a_k`
Out: number of divisible integers
Constraints: $n \le 10^{18}$; $k \le 20$; $2 \le a_i \le n$.

2187 – Bracket Sequences II

A prefix of a bracket sequence is fixed. Count valid bracket sequences of total length n that begin with exactly this prefix.
In: `n`; prefix string of length k
Out: number of completions modulo $10^9 + 7$
Constraints: $1 \le k \le n \le 10^6$.

2207 – Grundy's Game

Start with one heap of n coins. A move chooses a heap and splits it into two nonempty heaps of different sizes. Last move wins; determine the winner under optimal play. **In:** `t`; then t values n
Out: `first` or `second` for each test
Constraints: $t \le 10^5$; $n \le 10^6$.

2208 – Another Game

There are n nonempty heaps. A move selects some of the currently nonempty heaps and removes one coin from each selected heap. Last move wins; determine the winner. **In:** `t`; for each test: `n`; heap sizes
Out: `first` or `second`
Constraints: $t, n \le 2 \cdot 10^5$; $x_i \le 10^9$; total $n \le 2 \cdot 10^5$.

2209 – Counting Necklaces

Count colorings of a necklace with n pearls and m possible colors, where colorings equivalent by rotation are considered the same. **In:** `n m`
Out: number of distinct necklaces modulo $10^9 + 7$
Constraints: $1 \le n, m \le 10^6$.

2210 – Counting Grids

Count black/white colorings of an $n \times n$ grid, considering two grids equal if one can be rotated to obtain the other. **In:** `n`
Out: number of distinct grids modulo $10^9 + 7$
Constraints: $1 \le n \le 10^9$.

2417 – Counting Coprime Pairs

Given n positive integers, count unordered index pairs whose values are coprime, i.e. have gcd 1. **In:** `n`; `x_1 ... x_n`
Out: number of coprime pairs
Constraints: $n \le 10^5$; $x_i \le 10^6$.

3154 – System of Linear Equations

Solve n linear equations in m variables over the field modulo $10^9 + 7$: $\sum_j a_{i,j} x_j \equiv b_i \pmod{10^9 + 7}$. Any valid solution is accepted. **In:** `n m`; then n rows containing m coefficients and b_i
Out: $x_1, \dots, x_m$ with $0 \le x_i < 10^9 + 7$, or -1 if impossible
Constraints: $n, m \le 500$; all coefficients and b_i are in $[0, 10^9 + 7)$.

3355 – Sum of Four Squares

For each nonnegative integer n, find nonnegative integers a, b, c, d such that $n = a^2 + b^2 + c^2 + d^2$. **In:** `t`; then t values n
Out: one valid quadruple per test
Constraints: $t \le 1000$; $n \le 10^7$; sum of all $n \le 10^7$.

3396 – Next Prime

For each positive integer n, find the smallest prime strictly greater than n. **In:** `t`; then t values n
Out: next prime after each n
Constraints: $t \le 20$; $n \le 10^{12}$.

3397 – Permutation Order

Let $p(n, k)$ be the k-th permutation of $1 \dots n$ in lexicographic order. Support both directions: given (n, k) output $p(n, k)$, or given n and the permutation output its rank k. **In:** `t`; each test is `1 n k` or `2 n p_1 ... p_n`
Out: the requested permutation or rank
Constraints: $t \le 1000$; $n \le 20$; $1 \le k \le n!$.

3398 – Permutation Rounds

Start from sorted array $[1, 2, \dots, n]$. In each round, the element currently at position i moves to position p_i. Find the first positive number of rounds after which the array is sorted again. **In:** `n`; permutation `p_1 ... p_n`
Out: number of rounds modulo $10^9 + 7$
Constraints: $1 \le n \le 2 \cdot 10^5$.

3406 – Triangle Number Sums

Triangle numbers are $1 + 2 + \cdots + k$ for $k \ge 1$. For each positive integer n, find the minimum number of triangle numbers whose sum is n. **In:** `t`; then t values n
Out: minimum count for each test
Constraints: $t \le 100$; $n \le 10^{12}$.

CSES — String Algorithms Problem Reference

All tasks: 1 s, 512 MiB. Counts are modulo 10^9+7 where stated.

1731 — Word Combinations

Given lowercase string s and a dictionary of k distinct words, count the ordered ways to split all of s into dictionary words (words may be reused). **In:** s; k; then k words

Out: number of complete segmentations mod 10^9+7

Constraints: $|s| \le 5000$, $k \le 10^5$; total dictionary length $\le 10^6$.

1753 — String Matching

Given lowercase text s and pattern p, count starting positions where p occurs in s; overlapping occurrences count. **In:** s; p

Out: number of occurrences

Constraints: $1 \le |s|, |p| \le 10^6$.

1732 — Finding Borders

A border is a proper prefix that is also a suffix. Find the lengths of all borders of a lowercase string. **In:** s

Out: all border lengths in increasing order

Constraints: $1 \le |s| \le 10^6$.

1733 — Finding Periods

A period is a prefix whose repetitions generate the whole string; the final repetition may be partial. Find all valid period lengths (including $|s|$). **In:** s

Out: all period lengths in increasing order

Constraints: $1 \le |s| \le 10^6$.

1110 — Minimal Rotation

Among all cyclic rotations of a lowercase string, output the lexicographically smallest rotation. **In:** s

Out: lexicographically minimal rotation

Constraints: $1 \le |s| \le 10^6$.

1111 — Longest Palindrome

Find a longest palindromic *substring* of a lowercase string. If several exist, output any one. **In:** s

Out: any longest palindromic substring

Constraints: $1 \le |s| \le 10^6$.

3138 — All Palindromes

For every position i, find the maximum length of a palindromic substring that ends exactly at i. **In:** s

Out: n lengths, for positions $1, \dots, n$

Constraints: $1 \le n \le 2 \cdot 10^5$.

1112 — Required Substring

Count uppercase A–Z strings of length n that contain a given pattern p as a contiguous substring at least once. **In:** n; pattern p

Out: count mod 10^9+7

Constraints: $1 \le n \le 1000$, $1 \le |p| \le 100$.

2420 — Palindrome Queries

Maintain a lowercase string under: `1 k x` set $s_k = x$; `2 a b` ask whether $s[a..b]$ is a palindrome. **In:** n, m; s; then m operations

Out: `YES`/`NO` for each type 2 query

Constraints: $n, m \le 2 \cdot 10^5$.

2102 — Finding Patterns

Given a lowercase text and many lowercase patterns, decide independently whether each pattern occurs as a substring. **In:** s; k; then k patterns

Out: `YES` or `NO` for each pattern

Constraints: $|s| \le 10^5$, $k \le 5 \cdot 10^5$; total pattern length $\le 5 \cdot 10^5$.

2103 — Counting Patterns

For each of many patterns, count its starting positions in a fixed text; overlapping occurrences count. **In:** s; k; then k patterns

Out: occurrence count for each pattern

Constraints: $|s| \le 10^5$, $k \le 5 \cdot 10^5$; total pattern length $\le 5 \cdot 10^5$.

2104 — Pattern Positions

For each pattern, find the first 1-indexed position where it occurs in a fixed text, or -1 if absent. **In:** s; k; then k patterns

Out: first position or -1 for each pattern

Constraints: $|s| \le 10^5$, $k \le 5 \cdot 10^5$; total pattern length $\le 5 \cdot 10^5$.

2105 — Distinct Substrings

Count the distinct nonempty substrings of a lowercase string. **In:** s

Out: number of distinct substrings

Constraints: $1 \le |s| \le 10^5$; answer fits 64-bit.

1149 — Distinct Subsequences

Delete arbitrary characters without reordering the rest. Count the distinct *nonempty* strings obtainable as subsequences. **In:** s

Out: number of distinct nonempty subsequences mod 10^9+7

Constraints: $1 \le |s| \le 5 \cdot 10^5$.

2106 — Repeating Substring

Find a longest substring that occurs in at least two positions of s (occurrences may overlap). If none exists, print -1. **In:** s

Out: any longest repeating substring, or -1

Constraints: $1 \le |s| \le 10^5$.

2107 — String Functions

For every 1-indexed position i, compute $z(i)$ = longest prefix match starting at i (with $z(1) = 0$), and $\pi(i)$ = longest proper prefix ending at i. **In:** s

Out: one line of all z values; one line of all π values

Constraints: $1 \le |s| \le 10^6$.

3225 — Inverse Suffix Array

Given a permutation of $1, \dots, n$ claimed to be the lexicographic order of suffix starting positions, construct any lowercase a–z string having exactly that suffix array. **In:** n; suffix-array permutation

Out: any valid lowercase string; -1 if impossible

Constraints: $1 \le n \le 10^5$.

1113 — String Transform

Invert the Burrows–Wheeler transform defined by appending unique sentinel `#` (smaller than a–z), sorting all rotations, then taking their last characters. **In:** transformed string of length $n+1$, containing one `#`

Out: original lowercase string of length n

Constraints: $1 \le n \le 10^6$.

2108 — Substring Order I

Sort all *distinct* nonempty substrings lexicographically and output the kth one. **In:** s; k

Out: kth distinct substring

Constraints: $|s| \le 10^5$, $1 \le k \le n(n+1)/2$; k is guaranteed valid.

2109 — Substring Order II

Sort all substring *occurrences* lexicographically (duplicates appear with multiplicity) and output the kth one. **In:** s; k

Out: kth substring in the multiset order

Constraints: $|s| \le 10^5$, $1 \le k \le n(n+1)/2$.

2110 — Substring Distribution

For each length $L = 1, \dots, n$, count how many distinct substrings of s have length exactly L. **In:** s

Out: n counts for lengths $1, \dots, n$

Constraints: $1 \le |s| \le 10^5$.

CSES — Geometry Problem Reference

All tasks: 1 s, 512 MiB. Coordinates and answers are exact integers unless stated otherwise.

1740 — Intersection Points

Given n horizontal/vertical segments, count their intersection points. Parallel segments never intersect, and no segment endpoint is an intersection point. **In:** n; then endpoints $(x_1, y_1), (x_2, y_2)$

Out: number of intersections

Constraints: $n \le 10^5$; coordinates in $[-10^6, 10^6]$; each segment is nonempty.

1741 — Area of Rectangles

Given n axis-aligned rectangles, find the total area of their union; overlapping area is counted once. **In:** n; then opposite corners $(x_1, y_1), (x_2, y_2)$

Out: union area

Constraints: $n \le 10^5$; $-10^6 \le x_1 < x_2 \le 10^6$, same for y.

1742 — Robot Path

A robot starts at $(0, 0)$ and follows axis-aligned commands. It stops after all commands or immediately upon reaching any point visited earlier, possibly partway through a command. Find distance traveled until stopping. **In:** n; then direction $d \in \{U, D, L, R\}$ and distance x

Out: total distance traveled

Constraints: $n \le 10^5$, $1 \le x \le 10^6$.

2189 — Point Location Test

For directed line $p_1 \to p_2$, classify point p_3 as lying to its left, right, or on the line. **In:** t; then p_1, p_2, p_3 coordinates

Out: `LEFT`, `RIGHT`, or `TOUCH` per test

Constraints: $t \le 10^5$; coordinates in $[-10^9, 10^9]$; $p_1 \ne p_2$.

2190 — Line Segment Intersection

For two closed nondegenerate segments, determine whether they share at least one point. Endpoint touching and collinear overlap both count as intersection. **In:** t; then endpoints of both segments

Out: `YES`/`NO` per test

Constraints: $t \le 10^5$; coordinates in $[-10^9, 10^9]$.

2191 — Polygon Area

Given the vertices of a simple polygon in boundary order, compute its area exactly. **In:** n; then vertices (x_i, y_i)

Out: $2A$, where A is the polygon area

Constraints: $3 \le n \le 1000$; coordinates in $[-10^9, 10^9]$.

2192 — Point in Polygon

Given a simple polygon and m query points, classify each point as inside, outside, or on the polygon boundary. **In:** n, m; n polygon vertices; then m points

Out: `INSIDE`, `OUTSIDE`, or `BOUNDARY` per point

Constraints: $n, m \le 1000$; coordinates in $[-10^9, 10^9]$.

2193 — Polygon Lattice Points

For a simple polygon with integer-coordinate vertices, count lattice points strictly inside it and lattice points on its boundary. **In:** n; then integer vertices (x_i, y_i)

Out: two integers: interior count, boundary count

Constraints: $3 \le n \le 10^5$; coordinates in $[-10^9, 10^9]$.

2194 — Minimum Euclidean Distance

Among n distinct planar points, find the minimum Euclidean distance between two different points. **In:** n; then points (x, y)

Out: d^2 for the minimum distance d

Constraints: $2 \le n \le 2 \cdot 10^5$; coordinates in $[-10^9, 10^9]$.

2195 — Convex Hull

Find the convex hull of n distinct points. Output *all* input points lying on the hull boundary, including collinear points along hull edges; any output order is accepted. **In:** n; then points (x, y)

Out: k, then the k hull-boundary points

Constraints: $3 \le n \le 2 \cdot 10^5$; hull area is positive; coordinates in $[-10^9, 10^9]$.

3410 — Maximum Manhattan Distances

Points are inserted one by one into an initially empty set. After each insertion, report the maximum Manhattan distance $|x_1 - x_2| + |y_1 - y_2|$ between any two inserted points. **In:** n; then the points in insertion order

Out: one maximum distance after each insertion

Constraints: $n \le 2 \cdot 10^5$; points distinct; coordinates in $[-10^9, 10^9]$.

3411 — All Manhattan Distances

Given n distinct points, compute the sum of Manhattan distances over all unordered pairs. **In:** n; then points (x, y)

Out: $\sum_{i<j}(|x_i - x_j| + |y_i - y_j|)$

Constraints: $n \le 2 \cdot 10^5$; coordinates in $[-10^9, 10^9]$.

3427 — Line Segments Trace I

Each segment runs from $(0, y_1)$ to (m, y_2) and has integer slope. For every integer $x = 0, 1, \ldots, m$, find the maximum y attained by any segment there. **In:** n, m; then y_1, y_2 for each segment

Out: $m + 1$ maxima for $x = 0, \ldots, m$

Constraints: $n, m \le 10^5$, $0 \le y_1, y_2 \le 10^9$.

3428 — Line Segments Trace II

Segments have integer-coordinate endpoints, integer slope, and arbitrary active intervals $[x_1, x_2] \subseteq [0, m]$. For every integer x, find the maximum y among segments covering that x; output -1 if none does. **In:** n, m; then $(x_1, y_1), (x_2, y_2)$

Out: $m + 1$ values for $x = 0, \ldots, m$

Constraints: $n, m \le 10^5$; $0 \le x_1 < x_2 \le m$; $0 \le y_1, y_2 \le 10^9$.

3429 — Lines and Queries I

Process online queries: `1 a b` adds the infinite line $y = ax + b$; `2 x` asks the maximum value among all lines at that x. The first query is always an insertion. **In:** n; then n queries

Out: maximum value for each type 2 query

Constraints: $n \le 2 \cdot 10^5$; $|a|, |b| \le 10^9$; $0 \le x \le 10^5$.

3430 — Lines and Queries II

Process online queries: `1 a b l r` adds $y = ax + b$ active only for $x \in [l, r]$; `2 x` asks the maximum among lines active at x. **In:** n; then n queries

Out: maximum for each type 2 query, or `NO` if no line is active

Constraints: $n \le 2 \cdot 10^5$; $|a|, |b| \le 10^9$; $0 \le x, l, r \le 10^5$, $l \le r$.

CSES – Advanced Techniques Problem Reference

All tasks: 1 s, 512 MB. Counts are modulo $10^9 + 7$ only where explicitly stated.

1628 – Meet in the Middle

Given n positive integers, count subsets of the array elements whose sum is exactly x; equal values at different positions are distinct choices. **In:** `n x`; `t_1 ... t_n`
Out: number of subsets with sum x
Constraints: $n \le 40$; $x, t_i \le 10^9$. Use 64-bit count.

2136 – Hamming Distance

Given n binary strings of equal length k, find the minimum Hamming distance between any two distinct strings. **In:** `n k`; then n bit strings
Out: minimum number of differing positions
Constraints: $n \le 2 \cdot 10^4$; $k \le 30$.

3360 – Corner Subgrid Check

An $n \times n$ grid uses the first k uppercase letters. For each letter, determine whether some subgrid of height and width at least 2 has that letter in all four corners. **In:** `n k`; then n grid rows
Out: k lines: `YES`/`NO` for each letter
Constraints: $n \le 3000$; $k \le 26$.

2137 – Corner Subgrid Count

In an $n \times n$ binary grid, count subgrids of height and width at least 2 whose four corner cells are all black (`1`). **In:** `n`; then n binary rows
Out: number of beautiful subgrids
Constraints: $n \le 3000$. Use 64-bit count.

2138 – Reachable Nodes

Given a directed acyclic graph, for every node count how many nodes are reachable from it, including the node itself. **In:** `n m`; then m directed edges `a b`
Out: n reachability counts
Constraints: $n \le 5 \cdot 10^4$; $m \le 10^5$.

2143 – Reachability Queries

Given a directed graph, answer whether node b is reachable from node a for each query. **In:** `n m q`; m directed edges; then q pairs `a b`
Out: `YES` or `NO` per query
Constraints: $n \le 5 \cdot 10^4$; $m, q \le 10^5$.

2072 – Cut and Paste

Maintain a string under operations that cut the current substring at positions $[a, b]$, remove it, and append it to the end. Output the final string. **In:** `n m`; uppercase string; then m lines `a b`
Out: final string
Constraints: $n, m \le 2 \cdot 10^5$.

2073 – Substring Reversals

Maintain a string under operations that reverse the current substring at positions $[a, b]$. Output the final string. **In:** `n m`; uppercase string; then m lines `a b`
Out: final string
Constraints: $n, m \le 2 \cdot 10^5$.

2074 – Reversals and Sums

Maintain an integer array with operations: `1 a b` reverses subarray $[a, b]$; `2 a b` asks its sum. **In:** `n m`; array; then m operations
Out: sum for every type-2 operation
Constraints: $n \le 2 \cdot 10^5$; $m \le 10^5$; $0 \le x_i \le 10^9$. Use 64-bit sums.

2076 – Necessary Roads

A connected simple undirected graph is given. Find every road whose removal makes some pair of cities disconnected. **In:** `n m`; then m undirected edges
Out: number of necessary roads, then the roads in any order
Constraints: $n \le 10^5$; $m \le 2 \cdot 10^5$.

2077 – Necessary Cities

A connected simple undirected graph is given. Find every city whose removal, together with its incident roads, leaves some two other cities disconnected. **In:** `n m`; then m undirected edges
Out: number of necessary cities, then their list in any order
Constraints: $n \le 10^5$; $m \le 2 \cdot 10^5$.

2078 – Eulerian Subgraphs

Choose any subset of edges of an undirected graph while keeping all vertices. Count choices for which every vertex has even degree. **In:** `n m`; then m undirected edges
Out: number of Eulerian subgraphs modulo $10^9 + 7$
Constraints: $n \le 10^5$; $m \le 2 \cdot 10^5$.

2084 – Monster Game I

Process levels in order. At level $i < n$, kill or escape; level n must be killed. Killing costs $s_i \cdot f$ using the current skill factor f, then sets $f = f_i$; escaping changes nothing. Minimize total time. **In:** `n x`; strengths s_i; resulting skill factors f_i
Out: minimum total time
Constraints: $n \le 2 \cdot 10^5$; $s_i \le 10^6$ nondecreasing; $x \ge f_1 \ge \cdots \ge f_n \ge 1$. Use 64-bit.

CSES – Advanced Techniques Problem Reference

Continued. All tasks: 1 s, 512 MB.

2085 – Monster Game II

Same game as Monster Game I: kill/escape levels in order, final monster mandatory; killing i costs s_i times the current skill factor and then sets it to f_i. Here strengths and skill factors have no monotonicity guarantees. **In:** `n x`; strengths s_i; resulting factors f_i
Out: minimum total time
Constraints: $n \le 2 \cdot 10^5$; $x, s_i, f_i \le 10^6$. Use 64-bit.

2086 – Subarray Squares

Partition an array into exactly k nonempty contiguous subarrays. A part costs the square of the sum of its elements; minimize the total cost. **In:** `n k`; `x_1 ... x_n`
Out: minimum total cost
Constraints: $1 \le k \le n \le 3000$; $x_i \le 10^5$. Use 64-bit.

2087 – Houses and Schools

Houses are at positions $1 \dots n$ with c_i children at house i. Place k schools in houses; every child walks to a nearest school. Minimize total walking distance. **In:** `n k`; `c_1 ... c_n`
Out: minimum total distance
Constraints: $1 \le k \le n \le 3000$; $c_i \le 10^9$. Use 64-bit.

2088 – Knuth Division

Starting with the whole array as one subarray, repeatedly split one current subarray into two until all parts are singletons. Each split costs the sum of the values in the subarray being split. Minimize total cost. **In:** `n`; `x_1 ... x_n`
Out: minimum total split cost
Constraints: $n \le 5000$; $x_i \le 10^9$. Use 64-bit.

2111 – Apples and Bananas

There are n apples and m bananas, each with integer weight $1 \dots k$. For every total weight $w = 2 \dots 2k$, count pairs consisting of one apple and one banana with combined weight w. **In:** `k n m`; apple weights; banana weights
Out: $2k-1$ counts for totals $2 \dots 2k$
Constraints: $k, n, m \le 2 \cdot 10^5$. Use 64-bit counts.

2112 – One Bit Positions

Given a binary string, for every distance $d = 1 \dots n-1$, count pairs of positions i, j with $i - j = d$ and both characters equal to `1`. **In:** one binary string of length n
Out: $n-1$ counts, for distances $1 \dots n-1$
Constraints: $2 \le n \le 2 \cdot 10^5$.

2113 – Signal Processing

Slide a mask across a signal from left to right, including partial overlaps at both ends. At every alignment, output the sum of products of all aligned signal/mask values. **In:** `n m`; signal $a_1 \dots a_n$; mask $b_1 \dots b_m$
Out: $n+m-1$ overlap dot products in order
Constraints: $n, m \le 2 \cdot 10^5$; $a_i, b_i \le 100$. Use 64-bit sums.

2101 – New Roads Queries

Initially there are no roads; one undirected road is added each day in the given order. For each pair (a, b), find the first day when a and b become connected. **In:** `n m q`; m roads in build order; then q pairs
Out: earliest day for each query, or -1 if never connected
Constraints: $n, m, q \le 2 \cdot 10^5$.

2133 – Dynamic Connectivity

Maintain an undirected simple graph under edge insertions and deletions. Report the number of connected components before any event and after every event. **In:** `n m k`; m initial edges; then k events `t a b`
Out: $k+1$ component counts
Constraints: $n \le 10^5$; $m, k \le 10^5$; updates are always valid.

2121 – Parcel Delivery

A directed route $a \to b$ has capacity r parcels and cost c per parcel. Send exactly k parcels from city 1 to city n at minimum total cost. **In:** `n m k`; then routes `a b r c`
Out: minimum cost, or -1 if sending k parcels is impossible
Constraints: $n \le 500$; $m \le 1000$; $k \le 100$; $r, c \le 1000$.

2129 – Task Assignment

There are n employees and n tasks with cost c_{ij} for assigning task j to employee i. Assign every employee exactly one distinct task, minimizing total cost, and output one optimal assignment. **In:** `n`; then $n \times n$ cost matrix
Out: minimum cost; then n employee–task pairs
Constraints: $n \le 200$; $1 \le c_{ij} \le 1000$.

2130 – Distinct Routes II

A directed graph has teleporters that may each be used at most once over the entire game. Play exactly k days; each day starts at node 1 and must reach node n. Every teleporter use costs one coin. Minimize total coins and output the k routes. **In:** `n m k`; then m directed teleporters `a b`
Out: minimum total uses and k valid routes, or only -1 if impossible
Constraints: $n \le 500$; $m \le 1000$; $1 \le k \le n-1$.

CSES Sliding Window Problems — Problem Reference

All tasks: 1 s, 512 MiB. Windows are processed from left to right.

1076 — Sliding Window Median

For every length-k window of $x_1, \ldots, x_n$, find its median. If k is even, use the smaller of the two middle elements. **In:** `n k`; $x_1 \ \ldots \ x_n$
Out: $n - k + 1$ medians
Constraints: $1 \le k \le n \le 2 \cdot 10^5$, $1 \le x_i \le 10^9$.

1077 — Sliding Window Cost

For every length-k window, find the minimum total cost to make all elements equal, where changing u to v costs $|u - v|$.
In: `n k`; $x_1 \ \ldots \ x_n$
Out: $n - k + 1$ minimum costs
Constraints: $1 \le k \le n \le 2 \cdot 10^5$, $1 \le x_i \le 10^9$. Use 64-bit.

3219 — Sliding Window Mex

For every length-k window, find its mex: the smallest nonnegative integer not present in the window. **In:** `n k`; $x_1 \ \ldots \ x_n$
Out: $n - k + 1$ mex values
Constraints: $1 \le k \le n \le 2 \cdot 10^5$, $0 \le x_i \le 10^9$.

3220 — Sliding Window Sum

The array is generated online. For every length-k window compute its sum, then xor all window sums together. **In:** `n k`; `x a b c`
Generator: $x_1 = x$, $x_i = (ax_{i-1} + b) \bmod c$ for $2 \le i \le n$.
Out: xor of all window sums
Constraints: $1 \le k \le n \le 10^7$, $0 \le x, a, b \le 10^9$, $1 \le c \le 10^9$. Use 64-bit sums.

3221 — Sliding Window Minimum

The array is generated online. For every length-k window compute its minimum, then xor all window minimums together. **In:** `n k`; `x a b c`
Generator: $x_1 = x$, $x_i = (ax_{i-1} + b) \bmod c$ for $2 \le i \le n$.
Out: xor of all window minimums
Constraints: $1 \le k \le n \le 10^7$, $0 \le x, a, b \le 10^9$, $1 \le c \le 10^9$.

3222 — Sliding Window Distinct Values

For every length-k window, count how many distinct values it contains. **In:** `n k`; $x_1 \ \ldots \ x_n$
Out: $n - k + 1$ distinct-value counts
Constraints: $1 \le k \le n \le 2 \cdot 10^5$, $1 \le x_i \le 10^9$.

3223 — Sliding Window Inversions

For every length-k window, count inversions: pairs of positions $p < q$ inside the window with $x_p > x_q$. **In:** `n k`; $x_1 \ \ldots \ x_n$
Out: $n - k + 1$ inversion counts
Constraints: $1 \le k \le n \le 2 \cdot 10^5$, $1 \le x_i \le 10^9$. Use 64-bit.

3224 — Sliding Window Mode

For every length-k window, output its most frequent value. If several values have maximum frequency, choose the smallest.
In: `n k`; $x_1 \ \ldots \ x_n$
Out: $n - k + 1$ modes
Constraints: $1 \le k \le n \le 2 \cdot 10^5$, $1 \le x_i \le 10^9$.

3227 — Sliding Window Advertisement

A fence has n unit-width boards of heights x_i. For every k consecutive boards, find the largest axis-aligned rectangle that fits under those bars. **In:** `n k`; $x_1 \ \ldots \ x_n$
Out: $n - k + 1$ maximum rectangle areas
Constraints: $1 \le k \le n \le 2 \cdot 10^5$, $1 \le x_i \le 10^9$. Use 64-bit.

3405 — Sliding Window Or

The array is generated online. For every length-k window compute the bitwise OR of its elements, then xor all window ORs together. **In:** `n k`; `x a b c`
Generator: $x_1 = x$, $x_i = (ax_{i-1} + b) \bmod c$ for $2 \le i \le n$.
Out: xor of all window ORs
Constraints: $1 \le k \le n \le 10^7$, $0 \le x, a, b \le 10^9$, $1 \le c \le 10^9$.

3426 — Sliding Window Xor

The array is generated online. For every length-k window compute the bitwise XOR of its elements, then xor all window XORs together. **In:** `n k`; `x a b c`
Generator: $x_1 = x$, $x_i = (ax_{i-1} + b) \bmod c$ for $2 \le i \le n$.
Out: xor of all window XORs
Constraints: $1 \le k \le n \le 10^7$, $0 \le x, a, b \le 10^9$, $1 \le c \le 10^9$.

CSES – Interactive Problems Reference

All tasks: 1 s, 512 MB. Flush after every output line; respect the query budget and terminate immediately when finished.

3112 – Hidden Integer

A hidden integer x is fixed. Determine x by asking threshold questions of the form "is $y < x$?". **Start:** no initial input; begin querying immediately
Ask: print `? y`, $1 \le y \le 10^9$
Reply: `YES` iff $y < x$, otherwise `NO`
Finish: print `! x` and terminate
Constraints: $1 \le x \le 10^9$; at most 30 `?` queries.

3139 – Hidden Permutation

A hidden permutation $a_1, \ldots, a_n$ of $1, \ldots, n$ is fixed. Recover every value by comparing entries at chosen indices. **Start:** read n
Ask: print `? i j` to ask whether $a_i < a_j$
Reply: `YES` iff $a_i < a_j$, otherwise `NO`
Finish: print `! a_1 ... a_n` and terminate
Constraints: $n \le 1000$; at most 10^4 comparison queries.

3305 – K-th Highest Score

There are n distinct Finnish scores and n distinct Swedish scores. Within each country, query scores by descending rank; find the k-th highest score among all $2n$ coders. **Start:** read n, k
Ask: print `F i` or `S i` for that country's i-th highest score
Reply: the requested integer score
Finish: print `! s`, where s is the overall k-th highest score, then terminate
Constraints: $n \le 10^5$, $1 \le k \le 2n$; all $2n$ scores distinct, in $[1, 10^9]$; at most 100 score queries total.

3228 – Permuted Binary Strings

A hidden permutation $a_1, \ldots, a_n$ is fixed. For any chosen binary string $b_1 \ldots b_n$, the grader returns the string permuted as $b_{a_1} \ldots b_{a_n}$. Recover the permutation. **Start:** read n
Ask: print `? b`, where $b \in \{0,1\}^n$
Reply: the length-n string $b_{a_1} \ldots b_{a_n}$
Finish: print `! a_1 ... a_n` and terminate
Constraints: $n \le 1000$; at most 10 binary-string queries.

3273 – Colored Chairs

n chairs form a cycle; every chair is red or blue, and n is odd. Find an index i such that adjacent chairs i and $i+1$ have the same color, with $n+1 \equiv 1$. **Start:** read n
Ask: print `? i` to query chair i
Reply: `R` or `B`
Finish: print `! i` asserting chairs i and $i+1$ have equal color, then terminate
Constraints: $3 \le n \le 2 \cdot 10^5$, n odd; at most 20 color queries.

3140 – Inversion Sorting

A hidden permutation of $1, \ldots, n$ must be sorted. Each move reverses a chosen subarray; afterward the grader reports the permutation's new inversion count. You win as soon as it becomes 0. **Start:** read n
Ask: print `i j` to reverse positions $i \ldots j$
Reply: the inversion count after that reversal
Finish: when the returned count is 0, terminate immediately; there is no separate `!` command
Constraints: $n \le 1000$; at most $4n$ reversal operations.

CSES — Bitwise Operations Problem Reference

All tasks: 1 s, 512 MiB.

1146 — Counting Bits

For all integers $1 \le v \le n$, count the total number of set bits in their binary representations.

In: `n`

Out: total number of 1 bits over $1, \dots, n$

Constraints: $1 \le n \le 10^{15}$. Use 64-bit.

1655 — Maximum Xor Subarray

Given $x_1, \dots, x_n$, find the maximum bitwise XOR of any nonempty contiguous subarray.

In: `n`; `x_1 ... x_n`

Out: maximum subarray XOR

Constraints: $n \le 2 \cdot 10^5$, $0 \le x_i \le 10^9$.

3191 — Maximum Xor Subset

Choose any subset of the array and XOR its elements. Find the maximum obtainable XOR value.

In: `n`; `x_1 ... x_n`

Out: maximum subset XOR

Constraints: $n \le 2 \cdot 10^5$, $0 \le x_i \le 10^9$.

3211 — Number of Subset Xors

Consider XOR values of all subsets, including the empty subset with XOR 0. Count how many distinct values are obtainable.

In: `n`; `x_1 ... x_n`

Out: number of distinct subset XORs

Constraints: $n \le 2 \cdot 10^5$, $0 \le x_i \le 10^9$.

3192 — K Subset Xors

Consider the XOR of every one of the 2^n subsets, including the empty subset. Values are counted with multiplicity. Print the k smallest values in nondecreasing order.

In: `n k`; `x_1 ... x_n`

Out: k smallest subset XORs, including duplicates

Constraints: $n \le 2 \cdot 10^5$, $k \le \min(2^n, 2 \cdot 10^5)$, $0 \le x_i \le 10^9$.

3233 — All Subarray Xors

Find every distinct integer that occurs as the XOR of some nonempty contiguous subarray.

In: `n`; `x_1 ... x_n`

Out: first count k, then the k XOR values in increasing order

Constraints: $n \le 2 \cdot 10^5$, $0 \le x_i \le 10^6$.

2419 — Xor Pyramid Peak

The bottom row is $a_1, \dots, a_n$. Every cell above is XOR of its lower-left and lower-right cells. Find the single top cell.

In: `n`; `a_1 ... a_n`

Out: topmost value

Constraints: $n \le 2 \cdot 10^5$, $1 \le a_i \le 10^9$.

3194 — Xor Pyramid Diagonal

Build the same XOR pyramid. Print the *leftmost* value of every row, ordered from the bottom row up to the top.

In: `n`; `a_1 ... a_n`

Out: n leftmost row values, bottom $\to$ top

Constraints: $n \le 2 \cdot 10^5$, $1 \le a_i \le 10^9$.

3195 — Xor Pyramid Row

Build the XOR pyramid from the given bottom row. Print all values on the kth row *from the top*; that row has exactly k entries.

In: `n k`; `a_1 ... a_n`

Out: the k values on row k from the top

Constraints: $1 \le k \le n \le 2 \cdot 10^5$, $1 \le a_i \le 10^9$.

1654 — SOS Bit Problem

For every input element x, count array elements y satisfying each condition:

- $x \mid y = x$;
- $x \,\&\, y = x$;
- $x \,\&\, y \ne 0$.

Counts include occurrences, so equal array values are counted separately.

In: `n`; `x_1 ... x_n`

Out: three counts for each input element, in input order

Constraints: $n \le 2 \cdot 10^5$, $1 \le x_i \le 10^6$.

3141 — And Subset Count

For every $k = 0, 1, \dots, n$, count the nonempty subsets whose bitwise AND of all chosen elements is exactly k.

In: `n`; `a_1 ... a_n`

Out: $n+1$ counts for $k = 0, \dots, n$, modulo $10^9 + 7$

Constraints: $n \le 2 \cdot 10^5$, $0 \le a_i \le n$.

CSES — Construction Problems Reference

All tasks: 1 s, 512 MiB. Any valid construction is accepted unless stated otherwise.

2214 — Inverse Inversions

Construct a permutation p of $1, \ldots, n$ having exactly k inversions, where an inversion is a pair $i < j$ with $p_i > p_j$.

In: `n k`

Out: one valid permutation

Constraints: $1 \le n \le 10^6$, $0 \le k \le n(n-1)/2$.

2215 — Monotone Subsequences

For each test, construct a permutation of $1, \ldots, n$ whose *longest monotone subsequence* has exactly k elements. A monotone subsequence may be increasing or decreasing.

In: `t`; then `n k`

Out: a valid permutation for each test, or `IMPOSSIBLE`

Constraints: $t \le 1000$, $1 \le k \le n \le 100$.

3422 — Third Permutation

Given permutations a, b with $a_i \ne b_i$ for every position, construct a permutation c such that simultaneously $c_i \ne a_i$ and $c_i \ne b_i$ for every i.

In: `n`; permutation a; permutation b

Out: a valid permutation c, or `IMPOSSIBLE`

Constraints: $2 \le n \le 10^5$.

3423 — Permutation Prime Sums

Construct two permutations a, b of $1, \ldots, n$ such that $a_i + b_i$ is prime for every position i.

In: `n`

Out: the two permutations, or `IMPOSSIBLE`

Constraints: $1 \le n \le 10^5$.

1697 — Chess Tournament

Player i wants to play exactly x_i games. Choose unordered pairs of distinct players so every player gets the requested degree; each pair may play at most once.

In: `n`; `x_1 ... x_n`

Out: number k of games and the k player pairs; or `IMPOSSIBLE`

Constraints: $n \le 10^5$, $\sum_i x_i \le 2 \cdot 10^5$.

3424 — Distinct Sums Grid

Construct an $n \times n$ integer grid such that:

- every value $1, \ldots, n$ appears exactly n times overall;
- the n row sums and n column sums are all pairwise distinct, giving exactly $2n$ distinct sums.

In: `n`

Out: a valid grid, or `IMPOSSIBLE`

Constraints: $1 \le n \le 1000$.

2423 — Filling Trominos

Tile the entire $n \times m$ grid with L-trominos, each occupying exactly three cells in an L shape. If possible, output letters so two edge-adjacent cells have the same letter *iff* they belong to the same tromino.

In: `t`; then `n m`

Out: for each test, `YES` plus an n-row letter grid, or `NO`

Constraints: $t \le 100$, $1 \le n, m \le 100$.

2418 — Grid Path Construction

Given distinct squares $a = (y_1, x_1)$ and $b = (y_2, x_2)$ in an $n \times m$ grid, construct a path from a to b that visits *every grid square exactly once*. Moves are only U/D/L/R.

In: `t`; then `n m y_1 x_1 y_2 x_2`

Out: for each test, `YES` and a move string of length $nm - 1$, or `NO`

Constraints: $t \le 100$, $1 \le n, m \le 50$; endpoints are distinct.

CSES – Advanced Graph Problems Reference

All tasks: 1 s, 512 MB. Paths/walks and output witnesses are stated explicitly where relevant.

3303 – Nearest Shops

An undirected graph has anime shops in k cities. For every city, find the shortest distance to a shop in a *different* city; a local shop does not count for that city's answer. **In:** `n m k`; k shop cities; then m undirected edges
Out: n distances; -1 if no other shop is reachable
Constraints: $k \le n \le 10^5$; $m \le 2 \cdot 10^5$. Graph may be disconnected.

1134 – Prüfer Code

Decode a Prüfer code of length $n-2$ into its labeled tree. The code convention repeatedly removes the smallest-labeled current leaf and records its neighbor. **In:** `n`; then $n-2$ code values
Out: $n-1$ tree edges in any order
Constraints: $3 \le n \le 2 \cdot 10^5$; code values in $[1, n]$.

1702 – Tree Traversals

A binary tree has distinct labels. Given its preorder and inorder traversals, reconstruct and print the postorder traversal. **In:** `n`; preorder; inorder
Out: postorder traversal
Constraints: $n \le 10^5$; input is guaranteed to describe a valid binary tree.

1757 – Course Schedule II

Given prerequisite edges $a \to b$, output the valid course order that completes course 1 as early as possible, then course 2 as early as possible, and so on. This priority is by course number, not simply by the next available label. **In:** `n m`; then m prerequisites `a b`
Out: n courses in the required order
Constraints: $n \le 10^5$; $m \le 2 \cdot 10^5$; at least one valid schedule exists.

1756 – Acyclic Graph Edges

Orient every edge of an undirected graph so that the resulting directed graph has no directed cycle. **In:** `n m`; then m undirected edges
Out: m directed edges; any valid orientation
Constraints: $n \le 10^5$; $m \le 2 \cdot 10^5$.

2177 – Strongly Connected Edges

Orient every edge of a simple undirected graph so that the resulting directed graph is strongly connected. **In:** `n m`; then m undirected edges
Out: a valid orientation, or `IMPOSSIBLE`
Constraints: $n \le 10^5$; $m \le 2 \cdot 10^5$.

2179 – Even Outdegree Edges

Orient every edge of a simple undirected graph so that every vertex has even outdegree. **In:** `n m`; then m undirected edges
Out: a valid orientation, or `IMPOSSIBLE`
Constraints: $n \le 10^5$; $m \le 2 \cdot 10^5$.

1707 – Graph Girth

Find the length (number of edges) of the shortest cycle in a simple undirected graph. **In:** `n m`; then m undirected edges
Out: girth, or -1 if the graph is acyclic
Constraints: $n \le 2500$; $m \le 5000$.

3357 – Fixed Length Walk Queries

For each query (a, b, x) in a connected simple undirected graph, determine whether a *walk* of exactly x edges can start at a and end at b; vertices/edges may be revisited. **In:** `n m q`; edges; then queries `a b x`
Out: `YES/NO` per query
Constraints: $n \le 2500$; $m \le 5000$; $q \le 10^5$; $0 \le x \le 10^9$.

3111 – Transfer Speeds Sum

A weighted tree has edge transfer speeds. For vertices a, b, $d(a, b)$ is the minimum edge speed on their unique path. Sum $d(a, b)$ over all unordered vertex pairs. **In:** `n`; then $n-1$ edges `a b x`
Out: $\sum_{a<b} d(a, b)$
Constraints: $n \le 2 \cdot 10^5$; $x \le 10^6$. Use 64-bit.

3407 – MST Edge Check

For every edge of a connected weighted simple graph, determine whether it can belong to *some* minimum spanning tree. **In:** `n m`; then edges `a b w`
Out: `YES/NO` for each edge in input order
Constraints: $n \le 10^5$; $m \le 2 \cdot 10^5$; $w \le 10^9$.

3408 – MST Edge Set Check

For each queried set of edge indices, determine whether there exists one MST that contains *all* edges in that set simultaneously. **In:** `n m q`; weighted edges; then q edge-index sets
Out: `YES/NO` per set
Constraints: $n \le 10^5$; $m, q \le 2 \cdot 10^5$; total queried-set size $\le m$.

3409 – MST Edge Cost

For every edge, find the minimum possible spanning-tree cost under the condition that this edge must be included. **In:** `n m`; then edges `a b w`
Out: conditioned minimum cost for each edge in input order
Constraints: $n \le 10^5$; $m \le 2 \cdot 10^5$; $w \le 10^9$. Use 64-bit.

1677 – Network Breakdown

An undirected network starts with m edges; k specified connections break one by one. Report the number of connected components after each deletion. **In:** `n m k`; initial edges; then k broken edges
Out: k component counts, in breakdown order
Constraints: $n \le 10^5$; $m \le 2 \cdot 10^5$; $1 \le k \le m$.

CSES – Advanced Graph Problems Reference

Continued. All tasks: 1 s, 512 MB.

3114 – Tree Coin Collecting I

A tree has some coin vertices. For each (a, b), find the minimum length of a walk from a to b that visits *at least one* coin vertex; detours are allowed. **In:** `n q`; coin flags; tree edges; then q pairs `a b`
Out: minimum walk length per query
Constraints: $n, q \le 2 \cdot 10^5$; at least one coin exists.

3149 – Tree Coin Collecting II

A tree has some coin vertices. For each (a, b), find the minimum length of a walk from a to b that visits *every* coin vertex. **In:** `n q`; coin flags; tree edges; then q pairs `a b`
Out: minimum walk length per query
Constraints: $n, q \le 2 \cdot 10^5$; at least one coin exists. Use 64-bit if desired.

1700 – Tree Isomorphism I

For each test, compare two rooted trees of equal size. Node 1 is the root in both; determine whether the rooted tree shapes are isomorphic (labels otherwise irrelevant). **In:** `t`; per test: `n`, then edges of tree 1 and tree 2
Out: `YES`/`NO` per test
Constraints: sum of $n \le 10^5$; $2 \le n \le 10^5$.

1701 – Tree Isomorphism II

For each test, compare two *unrooted* trees of equal size and determine whether their shapes are isomorphic; vertex labels are irrelevant. **In:** `t`; per test: `n`, then edges of tree 1 and tree 2
Out: `YES`/`NO` per test
Constraints: sum of $n \le 10^5$; $2 \le n \le 10^5$.

1699 – Flight Route Requests

Initially there are no directed flights. Given reachability requests $a \rightsquigarrow b$, find the minimum number of one-way flight connections whose transitive reachability satisfies every request. **In:** `n m`; then m distinct requested pairs `a b`
Out: minimum number of flights needed
Constraints: $n \le 10^5$; $m \le 2 \cdot 10^5$.

1703 – Critical Cities

In a directed graph with a route from 1 to n, find all cities that occur on *every* route from 1 to n. **In:** `n m`; then m directed edges
Out: count, then critical cities in increasing order
Constraints: $n \le 10^5$; $m \le 2 \cdot 10^5$.

1203 – Visiting Cities

In a directed positive-weight graph, consider only minimum-price routes from city 1 to city n. Find all cities that occur on *every shortest-price route*. **In:** `n m`; then flights `a b c`
Out: count, then mandatory cities in increasing order
Constraints: $n \le 10^5$; $m \le 2 \cdot 10^5$; $c \le 10^9$; a route $1 \to n$ exists.

3308 – Graph Coloring

Color a simple graph using the minimum possible number of colors so adjacent vertices differ, and output an optimal coloring. **In:** `n m`; then m undirected edges
Out: chromatic number k, then n colors in $[1, k]$
Constraints: $n \le 16$; $m \le n(n-1)/2$.

3158 – Bus Companies

Company j sells a ticket of cost c_j that lets you travel directly between any two cities it serves. Starting from city 1, find the cheapest route cost to every city. **In:** `n m`; costs c_j; for each company, its number/list of served cities
Out: n minimum costs from city 1
Constraints: $n, m \le 10^5$; total company-city incidences $\le 2 \cdot 10^5$; $c_j \le 10^9$.

3358 – Split into Two Paths

Given a DAG, partition all vertices into exactly two vertex-disjoint directed paths so every vertex appears in exactly one path. One path may be empty; unused graph edges do not matter. **In:** `n m`; then m directed edges
Out: `YES` and the two path vertex lists, or `NO`
Constraints: $n \le 2 \cdot 10^5$; $m \le 5 \cdot 10^5$.

1704 – Network Renovation

A connected network is initially a tree. Add the minimum number of new undirected connections so the network remains connected after the failure of *any one* connection.
In: `n`; then $n-1$ tree edges
Out: minimum k, then k added edges; any optimum solution
Constraints: $3 \le n \le 10^5$.

1705 – Forbidden Cities

A connected undirected graph is fixed. For each query (a, b, c), determine whether there is a route from a to b that never visits forbidden city c. **In:** `n m q`; graph edges; then queries `a b c`
Out: `YES`/`NO` per query
Constraints: $n, q \le 10^5$; $m \le 2 \cdot 10^5$.

1752 – Creating Offices

In a tree, choose as many office vertices as possible so the distance between every pair of chosen vertices is at least d. **In:** `n d`; then $n-1$ tree edges
Out: maximum count, then any optimal set of cities
Constraints: $n, d \le 2 \cdot 10^5$.

1685 – New Flight Routes

Add the minimum number of directed flights to a directed graph so every city becomes reachable from every other city. **In:** `n m`; then m directed edges
Out: minimum k, then k added flights; any optimum solution
Constraints: $n \le 10^5$; $m \le 2 \cdot 10^5$.

CSES — Counting Problems Reference

All tasks: 1 s, 512 MiB. Unless stated otherwise, answers are modulo 10^9+7.

1075 — Counting Permutations

Count permutations p of $1,\dots,n$ such that no adjacent pair differs by 1, i.e. $|p_i - p_{i+1}| \neq 1$ for every i. **In:** `n`

Out: number of beautiful permutations mod 10^9+7

Constraints: $1 \leq n \leq 1000$.

1078 — Grid Paths II

In an $n \times n$ grid, move from $(1,1)$ to (n,n) using only right/down moves while avoiding m trapped cells. Start/end are free and traps are distinct. **In:** `n m`; then m traps `y x`

Out: number of valid paths mod 10^9+7

Constraints: $n \leq 10^6$, $m \leq 1000$, $1 \leq y,x \leq n$.

1080 — Empty String

Given a lowercase string, repeatedly remove any adjacent equal pair. Count the possible sequences of removals that delete the entire string. **In:** lowercase string s

Out: number of ways mod 10^9+7

Constraints: $1 \leq |s| \leq 500$.

2176 — Counting Bishops

Place exactly k bishops on an $n \times n$ chessboard so that no two share a diagonal. Count all placements. **In:** `n k`

Out: number of valid placements mod 10^9+7

Constraints: $1 \leq n \leq 500$, $1 \leq k \leq n^2$.

2228 — Counting Sequences

Count length-n sequences over $\{1,\dots,k\}$ in which every value $1,\dots,k$ appears at least once. **In:** `n k`

Out: number of sequences mod 10^9+7

Constraints: $1 \leq k \leq n \leq 10^6$.

2229 — Permutation Inversions

Count permutations of $1,\dots,n$ having exactly k inversions, where an inversion is a pair $i<j$ with $p_i > p_j$. **In:** `n k`

Out: number of permutations mod 10^9+7

Constraints: $1 \leq n \leq 500$, $0 \leq k \leq n(n-1)/2$.

2415 — Functional Graph Distribution

A functional graph has labeled nodes and outdegree exactly 1 at every node. For every $k=1,\dots,n$, count such graphs with exactly k connected components. **In:** `n`

Out: n lines: answer for each $k=1,\dots,n$ mod 10^9+7

Constraints: $1 \leq n \leq 5000$.

2421 — Counting Reorders

Reorder the multiset of characters of a lowercase string so that adjacent characters are never equal. Count distinct resulting strings. **In:** lowercase string s

Out: number of valid reorderings mod 10^9+7

Constraints: $1 \leq |s| \leq 5000$.

2429 — Grid Completion

Complete an $n \times n$ grid of `.`, `A`, `B` so every row and column contains exactly one `A` and exactly one `B`. Existing letters are fixed; initially each row/column has at most one of each. **In:** `n`; then n grid rows

Out: number of completions mod 10^9+7

Constraints: $2 \leq n \leq 500$.

3157 — Collecting Numbers Distribution

For a permutation of $1,\dots,n$, repeatedly scan left-to-right, collecting the next required values in increasing order until the scan ends. For every k, count permutations requiring exactly k scans. **In:** `n`

Out: n answers for $k=1,\dots,n$, mod 10^9+7

Constraints: $1 \leq n \leq 5000$.

3232 — Tournament Graph Distribution

A tournament has exactly one directed edge between each pair of labeled nodes. For every $k=1,\dots,n$, count tournaments with exactly k strongly connected components. **In:** `n`

Out: n lines: answer for each $k=1,\dots,n$ mod 10^9+7

Constraints: $1 \leq n \leq 500$.

3400 — Raab Game II

Each player owns cards $1,\dots,n$ and plays every card exactly once over n turns. Higher card scores 1 point, ties score neither; count ordered pairs of play orders giving final scores (a,b). **In:** `t`; then t tests `n a b`

Out: answer for each test mod 10^9+7

Constraints: $t \leq 1000$, $n \leq 5000$, $0 \leq a,b \leq n$.

3413 — Filled Subgrid Count I

Grid entries are the first k uppercase letters. For each letter, count contiguous axis-aligned *square* subgrids whose every cell equals that letter. **In:** `n k`; then n grid rows

Out: k exact counts, one per letter

Constraints: $n \leq 3000$, $k \leq 26$. Use 64-bit.

3414 — Filled Subgrid Count II

For each of the first k uppercase letters, count contiguous axis-aligned *rectangular* subgrids whose every cell equals that letter. **In:** `n k`; then n grid rows

Out: k exact counts, one per letter

Constraints: $n \leq 3000$, $k \leq 26$. Use 64-bit.

3415 — All Letter Subgrid Count I

Count contiguous axis-aligned *square* subgrids containing every one of the first k uppercase letters at least once. **In:** `n k`; then n grid rows

Out: exact number of qualifying squares

Constraints: $n \leq 3000$, $k \leq 26$. Use 64-bit.

3416 — All Letter Subgrid Count II

Count contiguous axis-aligned *rectangular* subgrids containing every one of the first k uppercase letters at least once. **In:** `n k`; then n grid rows

Out: exact number of qualifying rectangles

Constraints: $n \leq 500$, $k \leq 26$. Use 64-bit.

3417 — Border Subgrid Count I

For each letter, count contiguous axis-aligned *square* subgrids whose entire border consists of that letter; interior cells may be arbitrary. **In:** `n k`; then n grid rows

Out: k exact counts, one per letter

Constraints: $n \leq 3000$, $k \leq 26$. Use 64-bit.

3418 — Border Subgrid Count II

For each letter, count contiguous axis-aligned *rectangular* subgrids whose entire border consists of that letter; interior cells may be arbitrary. **In:** `n k`; then n grid rows

Out: k exact counts, one per letter

Constraints: $n \leq 500$, $k \leq 26$. Use 64-bit.

CSES — Additional Problems I Reference

All tasks: 1 s, 512 MiB. Modulo $10^9 + 7$ only where explicitly stated.

1086 — Writing Numbers

Write $1, 2, 3, \ldots$ in decimal order. Across the whole process, each digit key $0, \ldots, 9$ may be pressed at most n times; find the largest x for which every number $1, \ldots, x$ can be written completely. **In:** `n`

Out: last writable number x

Constraints: $1 \le n \le 10^{18}$. Use 64-bit.

1087 — Shortest Subsequence

Given a DNA string over `A,C,G,T`, find a shortest DNA string that is *not* a subsequence of it. Any shortest answer is accepted. **In:** DNA string s

Out: any shortest non-subsequence

Constraints: $1 \le |s| \le 10^6$.

1142 — Advertisement

A histogram has n unit-width bars of heights k_i. Find the maximum area of an axis-aligned rectangle contained under consecutive bars. **In:** `n`; $k_1 \ldots k_n$

Out: maximum rectangle area

Constraints: $n \le 2 \cdot 10^5$, $k_i \le 10^9$. Use 64-bit.

1147 — Maximum Building I

Given an $n \times m$ grid of empty cells `.` and trees `*`, find the largest axis-aligned rectangle consisting only of empty cells. **In:** `n m`; then n grid rows

Out: maximum empty-rectangle area

Constraints: $1 \le n, m \le 1000$.

1162 — Sorting Methods

Given a permutation, independently find the minimum operations needed to sort it using: (1) adjacent swaps; (2) arbitrary swaps; (3) remove one element and insert it anywhere; (4) remove one element and insert it at the front. **In:** `n`; permutation p

Out: four minima in the order above

Constraints: $n \le 2 \cdot 10^5$. First answer needs 64-bit.

1188 — Bit Inversions

Maintain a bit string under m point flips. After each flip, report the maximum length of a contiguous run consisting entirely of equal bits. **In:** bit string s; `m`; flip positions $x_1 \ldots x_m$

Out: m run lengths, after each flip

Constraints: $|s|, m \le 2 \cdot 10^5$.

1191 — Cyclic Array

Partition a positive cyclic array into the minimum number of nonempty contiguous cyclic segments such that every segment sum is at most k. A feasible partition is guaranteed. **In:** `n k`; $x_1 \ldots x_n$

Out: minimum number of segments

Constraints: $n \le 2 \cdot 10^5$, $x_i \le 10^9$, $k \le 10^{18}$; $x_i \le k$.

1670 — Swap Game

A 3×3 grid contains $1, \ldots, 9$ once each. Swap horizontally/vertically adjacent cells; find the minimum swaps to reach row-major order $1, 2, \ldots, 9$. **In:** three rows of three integers

Out: minimum number of adjacent swaps

1747 — Pyramid Array

Given n distinct values, use adjacent swaps to obtain an array that is first strictly increasing and then strictly decreasing; either part may be empty, so a fully monotone array is allowed. Minimize swaps. **In:** `n`; $x_1 \ldots x_n$

Out: minimum adjacent swaps

Constraints: $n \le 2 \cdot 10^5$, $x_i \le 10^9$. Use 64-bit.

2186 — Special Substrings

Let C be the set of characters occurring anywhere in the input string. Count substrings in which every character in C occurs the same number of times. **In:** lowercase string s

Out: number of special substrings

Constraints: $|s| \le 2 \cdot 10^5$. Use 64-bit.

2414 — List of Sums

An unknown list A has n positive integers. The input multiset B contains every unordered pair sum $a_i + a_j$ for $i < j$; reconstruct any valid A. Existence is guaranteed. **In:** `n`; then $n(n-1)/2$ pair sums

Out: any valid n values of A, in any order

Constraints: $3 \le n \le 100$; values of A are positive and at most 10^9.

2422 — Multiplication Table

For odd n, consider the multiset $\{ij : 1 \le i, j \le n\}$ of the $n \times n$ multiplication table. Find its middle element after sorting all n^2 entries. **In:** `n`

Out: median table value

Constraints: $1 \le n < 10^6$, n odd. Use 64-bit arithmetic.

2425 — Stack Weights

Coins satisfy unknown strictly increasing weights $w_1 < w_2 < \cdots < w_n$. Process placements into left/right stacks; after each, determine whether the left stack is certainly heavier, the right is certainly heavier, or neither conclusion follows from the ordering alone. **In:** `n`; then n moves `c s`, $s \in \{1, 2\}$

Out: after each move: `>` left, `<` right, or `?`

Constraints: $n \le 2 \cdot 10^5$.

2427 — Letter Pair Move Game

A length-$2n$ string has exactly $n-1$ `A`'s, $n-1$ `B`'s, and one adjacent pair of empty boxes. A move transfers any adjacent occupied pair into the current empty pair, preserving order. Reach a state with every `A` before every `B`. **In:** `n`; initial string over `A,B,.`

Out: any valid solution with $k \le 1000$ moves, or `-1` if impossible

Constraints: $1 \le n \le 100$; if solvable, a solution of at most $10n$ moves exists.

3150 — Distinct Values Sum

For every subarray $x_a, \ldots, x_b$, let $d(a, b)$ be its number of distinct values. Compute $\sum_{a \le b} d(a, b)$ over all subarrays. **In:** `n`; $x_1 \ldots x_n$

Out: the total distinct-value sum

Constraints: $n \le 2 \cdot 10^5$, $x_i \le 10^9$. Use 64-bit.

CSES — Additional Problems I Reference

Continued. Modulo 10^9+7 only where explicitly stated.

3151 — Bubble Sort Rounds I

Bubble sort scans left-to-right each round, swapping adjacent out-of-order pairs as encountered. Given an arbitrary integer array, determine how many rounds are needed until it is sorted. **In:** `n`; $x_1 \dots x_n$

Out: number of bubble-sort rounds

Constraints: $n \le 2 \cdot 10^5$, $x_i \le 10^9$.

3152 — Bubble Sort Rounds II

Using the same left-to-right bubble-sort round, find the entire array after exactly k rounds; k may be much larger than n. **In:** `n k`; $x_1 \dots x_n$

Out: array after k rounds

Constraints: $n \le 2 \cdot 10^5$, $0 \le k \le 10^9$, $x_i \le 10^9$.

3169 — Counting LCM Arrays

For each test, count positive-integer arrays $a_1, \dots, a_n$ such that $\mathrm{lcm}(a_i, a_{i+1}) = k$ for every adjacent pair. **In:** `t`; then tests `n k`

Out: answer for each test mod 10^9+7

Constraints: $t \le 1000$, $2 \le n \le 10^9$, $1 \le k \le 10^9$.

3175 — Beautiful Permutation II

Construct the lexicographically smallest permutation of $1, \dots, n$ such that no adjacent values differ by 1. If none exists, report failure. **In:** `n`

Out: lexicographically minimum beautiful permutation, or `NO SOLUTION`

Constraints: $1 \le n \le 10^6$.

3190 — Distinct Values Splits

Split the whole array into one or more contiguous segments so that every segment contains pairwise distinct values. Count all possible choices of cut positions. **In:** `n`; $x_1 \dots x_n$

Out: number of valid splits mod 10^9+7

Constraints: $n \le 2 \cdot 10^5$, $x_i \le 10^9$.

3193 — Square Subsets

Count subsets of the given multiset whose element product is a perfect square. The empty subset is included and has product 1. **In:** `n`; $x_1 \dots x_n$

Out: number of square-product subsets mod 10^9+7

Constraints: $n \le 5000$, $1 \le x_i \le 5000$.

3213 — Water Containers Moves

Containers have capacities a, b and start empty. A move fills one container, empties one, or pours between them until the source empties or destination fills. End with exactly x units in A, minimizing the *total amount of water moved* across all moves. **In:** `a b x`

Out: minimum-water valid move sequence and its total, or `-1`

Constraints: $1 \le a, b, x \le 1000$; every printed move must transfer at least one unit.

3214 — Water Containers Queries

For many tests with capacities a, b, starting from both empty and using the same fill/empty/full-pour operations, decide whether it is possible to finish with exactly x units in container A. **In:** `t`; then tests `a b x`

Out: `YES` or `NO` for each test

Constraints: $t \le 1000$, $a, b, x \le 10^9$.

3294 — Subarray Sum Constraints

Construct integers $x_1, \dots, x_n$ satisfying every constraint (l, r, s): $x_l + \cdots + x_r = s$. Output any solution with $|x_i| \le 10^{15}$, or detect inconsistency. **In:** `n m`; then m constraints `l r s`

Out: `YES` and any valid array, or `NO`

Constraints: $n \le 5000$, $m \le 2 \cdot 10^5$, $|s| \le 10^9$. Use 64-bit.

3301 — Maximum Average Subarrays

For every endpoint i, among all nonempty subarrays ending at i, choose one with maximum average; if several tie, choose the longest. Report its length. **In:** `n`; $x_1 \dots x_n$

Out: n optimal lengths, one for each endpoint

Constraints: $n \le 2 \cdot 10^5$, $1 \le x_i \le 10^6$.

3302 — Subsets with Fixed Average

Count nonempty subsets whose arithmetic mean is exactly the given integer a. **In:** `n a`; $x_1 \dots x_n$

Out: number of qualifying subsets mod 10^9+7

Constraints: $n \le 500$, $a, x_i \le 500$.

3306 — Nearest Campsites I

Given n reserved and m free grid points, distance is Manhattan distance. For each free point consider its nearest reserved point; output the maximum of these nearest distances. **In:** `n m`; n reserved points; m free points

Out: $\max_{q \in F} \min_{p \in R} \|p - q\|_1$

Constraints: $n, m \le 10^5$, coordinates $\le 10^6$.

3307 — Nearest Campsites II

For every free campsite, compute its Manhattan distance to the nearest reserved campsite, preserving the input order of free points. **In:** `n m`; n reserved points; m free points

Out: m nearest-reserved distances

Constraints: $n, m \le 10^5$, coordinates $\le 10^6$.

3361 — Two Array Average

Choose a nonempty prefix of each of two length-n arrays. Maximize the average of all selected elements combined, and output any pair of prefix lengths attaining the optimum. **In:** `n`; array a; array b

Out: optimal prefix sizes i, j

Constraints: $n \le 10^5$, $a_i, b_i \le 10^9$; checker tolerance 10^{-6} on achieved optimum.

3404 — Permutation Subsequence

Given two arrays that are permutations, find a longest common subsequence: values must appear in the same left-to-right order in both arrays, with gaps allowed. **In:** `n m`; permutation a; permutation b

Out: LCS length, then any corresponding sequence

Constraints: $n, m \le 2 \cdot 10^5$; $1 \le a_i \le n$, $1 \le b_i \le m$.

CSES — Additional Problems II Reference

All tasks: 1 s, 512 MiB. Modulo 10^9+7 only where explicitly stated.

1148 — Maximum Building II

Given an $n \times m$ forest grid of empty cells `.` and trees `*`, count placements of an all-empty axis-aligned rectangle for *every exact size* $h \times w$. **In:** `n m`; then n grid rows

Out: n rows; row h contains the counts for widths $w = 1, \dots, m$

Constraints: $1 \le n, m \le 1000$.

1157 — Mex Grid Queries

An infinite positive-index grid is filled so cell (y, x) is the smallest nonnegative integer absent both to its left in row y and above it in column x. Find the value at one queried cell. **In:** `y x`

Out: value at (y, x)

Constraints: $1 \le y, x \le 10^9$.

1159 — Book Shop II

Book i costs h_i, gives s_i pages, and has at most k_i copies available. With total cost at most x, maximize purchased pages. **In:** `n x`; prices h; pages s; copy limits k

Out: maximum total pages

Constraints: $n \le 100$, $x \le 10^5$, $h_i, s_i, k_i \le 1000$.

1161 — Stick Divisions

Start with one stick of length x. A move splits one current stick into two and costs the original stick length; end with the multiset of target lengths $d_1, \dots, d_n$. Minimize total cost. **In:** `x n`; $d_1 \dots d_n$

Out: minimum total split cost

Constraints: $x \le 10^9$, $n \le 2 \cdot 10^5$, $\sum d_i = x$. Use 64-bit.

1189 — Food Division

Children form a cycle. Child i initially has a_i food and must end with b_i; one step transfers one unit to either neighbor. The total food is already correct. Minimize transfers. **In:** `n`; $a_1 \dots a_n$; $b_1 \dots b_n$

Out: minimum number of unit transfers

Constraints: $n \le 2 \cdot 10^5$, $a_i, b_i \le 10^6$. Use 64-bit.

1665 — Coding Company

Partition the n coders into teams. A team's penalty is $\max t - \min t$; count divisions whose sum of team penalties is at most x. **In:** `n x`; skills $t_1 \dots t_n$

Out: number of valid divisions mod 10^9+7

Constraints: $n \le 100$, $x \le 5000$, $0 \le t_i \le 100$.

1698 — Swap Round Sorting

Sort a permutation in the minimum number of *swap rounds*; within one round, choose any number of distinct pairs and swap each chosen pair. Also construct the swaps in every round. **In:** `n`; permutation p

Out: minimum round count; for each round, its swaps as index pairs

Constraints: $n \le 2 \cdot 10^5$.

1706 — School Excursion

Each given pair of children insists on visiting the same attraction, so every connected component must stay together. For each $i = 1, \dots, n$, decide whether exactly i children can be assigned to Korkeasaari. **In:** `n m`; then m same-attraction pairs

Out: length-n bit string; bit i is 1 iff size i is possible

Constraints: $n, m \le 10^5$.

1709 — Coin Grid

An $n \times n$ binary coin grid can be cleared by a move that removes all remaining coins in one chosen row or column. Find the minimum number of moves and output any optimal set of rows/columns. **In:** `n`; then n rows over `.,o`

Out: minimum k, then k moves `1 r` (row) or `2 c` (column)

Constraints: $1 \le n \le 100$.

2075 — Reversal Sorting

Sort a permutation by reversing contiguous subarrays. Any construction using at most n reversals is accepted; the number of reversals need not be minimum. **In:** `n`; permutation p

Out: $k \le n$, then reversal intervals $[a, b]$

Constraints: $n \le 2 \cdot 10^5$.

2115 — Bit Substrings

For each $k = 0, 1, \dots, n$, count the *nonempty* substrings of a bit string that contain exactly k ones. **In:** binary string s

Out: $n+1$ counts for $k = 0, \dots, n$

Constraints: $|s| \le 2 \cdot 10^5$. Counts need 64-bit.

2131 — Grid Puzzle II

Choose grid cells so row i contains exactly a_i chosen cells and column j exactly b_j. Cell (i, j) yields c_{ij} coins; maximize the collected sum and construct the chosen cells, or detect infeasibility. **In:** `n`; row quotas a; column quotas b; then coin matrix c

Out: maximum sum and an `X/.` grid, or `-1`

Constraints: $n \le 50$, $a_i, b_j \le n$, $c_{ij} \le 1000$; $\sum a_i = \sum b_j$.

2132 — Increasing Array II

Change each array value by unit increments/decrements until the sequence is nondecreasing. Minimize the total absolute change. **In:** `n`; $x_1 \dots x_n$

Out: minimum number of unit changes

Constraints: $n \le 2 \cdot 10^5$, $x_i \le 10^9$. Use 64-bit.

2174 — Removing Digits II

Starting from n, repeatedly subtract any one decimal digit currently appearing in the number. Find the minimum steps to reach 0. **In:** `n`

Out: minimum number of steps

Constraints: $1 \le n \le 10^{18}$. Use 64-bit.

2180 — Coin Arrangement

A $2 \times n$ grid contains exactly $2n$ coins total, possibly multiple per cell. Move individual coins one orthogonal step per operation until every cell has exactly one; minimize total moves. **In:** `n`; then the two rows of coin counts

Out: minimum number of moves

Constraints: $1 \le n \le 10^5$. Use 64-bit.

CSES — Additional Problems II Reference

Continued. Modulo $10^9 + 7$ only where explicitly stated.

2402 — Two Stacks Sorting

Process a permutation from left to right using two stacks: move the first remaining input value onto one stack, or pop a stack value to the output. The final output must be $1, 2, \ldots, n$. **In:** `n`; input permutation

Out: for each input number, a chosen stack `1` or `2`; or `IMPOSSIBLE`

Constraints: $n \le 2 \cdot 10^5$.

2426 — Programmers and Artists

Each applicant has programming skill x and artistic skill y. Hire exactly a programmers and b artists, with no applicant used twice, maximizing total skill. **In:** `a b n`; then n pairs `x y`

Out: maximum total skill

Constraints: $n \le 2 \cdot 10^5$, $a + b \le n$, $x, y \le 10^9$. Use 64-bit.

2430 — Binary Subsequences

Construct a shortest bit string having exactly n distinct *nonempty* subsequences (identical resulting strings count once). Any minimum-length construction is accepted. **In:** `n`

Out: any minimum-length valid bit string

Constraints: $1 \le n \le 10^6$.

2432 — Grid Puzzle I

Choose cells of an $n \times n$ grid so row i has exactly a_i chosen cells and column j exactly b_j. Construct any feasible `X/.` grid. **In:** `n`; row quotas a; column quotas b

Out: a valid grid, or `-1` if impossible

Constraints: $n \le 50$, $0 \le a_i, b_j \le n$.

3108 — K Subset Sums I

Consider all 2^n subsets, including the empty subset. Output the k smallest subset sums in nondecreasing order; equal sums from different subsets appear with multiplicity. **In:** `n k`; $x_1 \ldots x_n$

Out: k smallest subset sums

Constraints: $n \le 2 \cdot 10^5$, $k \le 2 \cdot 10^5$, $|x_i| \le 10^9$. Use 64-bit sums.

3109 — K Subset Sums II

Consider all subsets containing exactly m array elements. Output the k smallest sums in nondecreasing order, retaining multiplicity between different subsets. **In:** `n m k`; $x_1 \ldots x_n$

Out: k smallest size-m subset sums

Constraints: $m < n \le 2 \cdot 10^5$, $k \le 2 \cdot 10^5$, $|x_i| \le 10^9$. Use 64-bit sums.

3159 — Replace with Difference

Repeatedly choose two *current numbers* a, b, delete both, and insert $|a - b|$. Construct $n - 1$ operations so the sole remaining value is 0, or report impossibility. **In:** `n`; $x_1 \ldots x_n$

Out: chosen value pairs a, b for every operation, or `-1`

Constraints: $2 \le n \le 1000$, $1 \le x_i \le 1000$.

3161 — GCD Subsets

For every $k = 1, \ldots, n$, count nonempty subsets of array positions whose selected values have greatest common divisor exactly k. **In:** `n`; $x_1 \ldots x_n$

Out: n answers for gcd $1, \ldots, n$, modulo $10^9 + 7$

Constraints: $n \le 2 \cdot 10^5$, $1 \le x_i \le n$.

3215 — Bouncing Ball Steps

A ball starts at $(1, 1)$ in an $n \times m$ grid moving diagonally down-right, one cell per step, reflecting whenever it hits a border. For a huge k, find its position and total number of direction changes after k steps. **In:** `t`; then tests `n m k`

Out: for each test: row, column, direction-change count

Constraints: $t \le 1000$, $n, m \le 10^9$, $k \le 10^{18}$. Use 64-bit.

3216 — Bouncing Ball Cycle

For the same reflecting diagonal motion, determine when the ball first reaches $(1, 1)$ again and how many distinct grid cells were visited during the cycle. **In:** `t`; then tests `n m`

Out: for each test: return time and number of distinct visited cells

Constraints: $t \le 1000$, $2 \le n, m \le 10^9$. Use 64-bit.

3218 — Knight Moves Queries

On the infinite chessboard with positive row/column indices, answer the minimum knight moves from each queried (x, y) to the top-left corner $(1, 1)$. **In:** `q`; then q positions `x y`

Out: minimum knight distance for each query

Constraints: $q \le 10^5$, $x, y \le 10^9$.

3312 — Grid Coloring II

Every cell initially contains `A`, `B`, or `C`. Change *every* cell to a different letter so that no orthogonally adjacent final cells are equal. Construct any solution. **In:** `n m`; then n grid rows

Out: a valid recolored grid, or `IMPOSSIBLE`

Constraints: $n, m \le 500$.

3401 — Stick Difference

Make exactly k cuts among integer-length sticks, producing $n + k$ positive-integer pieces. For every $k = 1, \ldots, m$, minimize $\max(\text{piece}) - \min(\text{piece})$. **In:** `n m`; lengths $a_1 \ldots a_n$

Out: m minima for exactly $k = 1, \ldots, m$ cuts

Constraints: $n \le 10^5$, $m \le 2 \cdot 10^5$, $a_i \le 10^9$; all m cuts are feasible.

3402 — Minimum Cost Pairs

For each $k = 1, \ldots, \lfloor n/2 \rfloor$, choose k disjoint pairs of array elements. Pair (a, b) costs $|a - b|$; minimize the total cost. **In:** `n`; $x_1 \ldots x_n$

Out: $\lfloor n/2 \rfloor$ minimum costs

Constraints: $n \le 2 \cdot 10^5$, $x_i \le 10^9$. Use 64-bit.

3425 — Same Sum Subsets

Choose two disjoint nonempty subsets of the given positive elements whose sums are equal. Output any valid pair of subsets. **In:** `n`; $x_1 \ldots x_n$

Out: for each subset, its size and elements; or `IMPOSSIBLE`

Constraints: $3 \le n \le 40$, $\sum x_i \le 2^n - 2$.